

# ClayQuant

Calculation of X-ray powder diffraction patterns of interstratified clay minerals, and quantification of clay mineral assemblages from oriented mounts

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**Scope.** ClayQuant computes one-dimensional ( $00l$ ) and three-dimensional ( $hkl$ ) X-ray powder diffraction patterns for mica-type (illite), chlorite and kaolinite layer structures, for the ethylene glycol–smectite complex, and for randomly or non-randomly interstratified two-component stacks of these layers, with arbitrary preferred orientation and crystallite thickness distribution. It then uses those calculated patterns to quantify the clay mineral assemblage of a clay separate measured as three oriented mounts (air-dried, ethylene glycol-solvated and heated), by non-negative least squares over the whole pattern.

The program is written in Python, is operated through a local browser interface or from the command line, and is distributed as source. This manual documents the physical model, the implementation, the operating procedure, every control of the interface, the output files, and the tests by which the program was verified.

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# 1 Introduction

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## 1.1 Purpose

Quantitative analysis of clay mineral assemblages by X-ray diffraction is made difficult by three properties of the minerals themselves. Clay minerals are platy, so a specimen prepared for analysis is strongly textured rather than randomly oriented. They are frequently interstratified, so that a crystallite is a stack of two or more different layer types and produces neither the pattern of one end member nor a superposition of both. And they are finely divided, so that crystallites are only a few tens of layers thick and their reflections are correspondingly broad. None of these is a defect of the measurement; all three are properties of the material that a calculation must reproduce before a quantification can mean anything.

ClayQuant addresses these by calculating the pattern that each candidate phase would produce *under the conditions of the measurement* — its texture, its crystallite thickness, its layer spacing, the instrument's emission profile and slit geometry — and then fitting the measured pattern as a non-negative combination of those calculated patterns.

## 1.2 What the program does and does not do

It does:

- compute  $00l$  and  $hkl$  intensities from crystal structures supplied as CIF;
- compute the diffracted intensity of a two-component interstratified stack with arbitrary junction probabilities and a distribution of crystallite thicknesses;
- apply preferred orientation, Lorentz–polarization, beam overflow, an arbitrary emission profile and an adjustable peak shape;
- read the common vendor file formats, determine the  $2\theta$  zero error, model the background, identify kaolinite and expandable clay from the three-mount comparison, identify accompanying non-clay minerals, and fit and report the assemblage.

It does not:

- refine structures. Cell parameters, atomic coordinates, site occupancies and displacement parameters are taken as given and are not adjusted against the data. ClayQuant is not a Rietveld program;
- refine peak positions, widths or orientation continuously. The fit chooses among *pre-computed* patterns, so the achievable agreement is limited by how finely the library samples those parameters (§3.3, §9);
- report weight percent. What is reported is each phase's share of diffracted intensity; the conversion requires a reference intensity ratio per phase (§2.13);
- model three-component interstratifications, or interstratifications with a lateral ( $ab$ ) component of disorder beyond that implied by the treatment of §2.12.

## 1.3 Notation and conventions

Throughout,  $\lambda$  is the wavelength in ångström,  $\theta$  the Bragg angle, and

$$s = 2 \sin\theta / \lambda = 1/d \tag{1.1}$$

the length of the scattering vector, in  $\text{\AA}^{-1}$ . Note that  $s$  is defined here without the factor  $2\pi$  and without the factor  $\frac{1}{2}$  used in some crystallographic literature; the quantity  $q = \sin\theta/\lambda = s/2$  is used only as the argument of scattering factors and displacement factors, matching International Tables convention. The stacking axis of a layer structure is  $c^*$ , and heights along it are given in  $\text{\AA}$  and denoted  $z$ . Layer repeat distances are denoted  $d_A, d_B$ . Intensities are per unit incident flux and are reported in arbitrary units, since only relative intensities enter the analysis.

## 1.4 Related programs

The one-dimensional part of the calculation follows the same physics as NEWMOD (Reynolds), PyXRD (Dumon & Van Ranst) and Sybilla, all of which implement the matrix formalism of §2.5. ClayQuant differs from these chiefly in coupling the calculation directly to a whole-pattern non-negative least-squares quantification, in computing the three-dimensional  $hkl$  contribution of the host phases from crystal structures rather than treating only basal reflections, and in being open source and scriptable.

# 2 Theory

## 2.1 The oriented mount

A clay separate is deposited on a glass slide so that the platelets lie with (001) close to the slide surface. In Bragg–Brentano reflection geometry the diffraction vector is always parallel to the specimen normal, so such a specimen presents predominantly its basal (00 $l$ ) reflections. The degree to which this is true is a property of the preparation, not a constant, and is treated here as an adjustable texture parameter (§2.9).

Three mounts of the same separate are measured:

| Mount           | Treatment                                      | Diagnostic role                                                                          |
|-----------------|------------------------------------------------|------------------------------------------------------------------------------------------|
| Air-dried       | as deposited, equilibrated with laboratory air | reference state                                                                          |
| Glycol-solvated | exposed to ethylene glycol vapour              | expandable layers swell to a two-layer complex near 16.9 $\text{\AA}$                    |
| Heated          | typically 500 $^{\circ}\text{C}$               | kaolinite dehydroxylates and its 7.15 $\text{\AA}$ reflection is lost; chlorite survives |

The quantitative fit is performed on the glycol-solvated mount, because glycolation separates the expandable from the non-expandable component and resolves the illite/smectite series, which in the air-dried state overlap near 10–14  $\text{\AA}$ .

## 2.2 Scattering factors and the structure factor

Atomic scattering factors  $f_0(q)$  are evaluated from the eleven-parameter analytical expansion of Waasmaier & Kirfel (1995),

$$f_0(q) = c + \sum_{i=1}^5 a_i \exp(-b_i q^2)$$

(2.1)

valid to  $q = 6 \text{ \AA}^{-1}$ , with  $q = \sin\theta/\lambda$ . Ionic species are used where the tabulation provides them and the CIF specifies a charge; otherwise the neutral atom is used, and a species that is not tabulated falls back to its neutral atom with a warning. Isotropic displacement is applied as

$$T(q) = \exp(-B q^2) \quad (2.2)$$

with  $B$  in  $\text{\AA}^2$ . Where a CIF reports  $U_{\text{iso}}$ , the conversion  $B = 8\pi^2 U$  is applied. The structure factor of a reflection  $hkl$  is then

$$F(hkl) = \sum_j n_j f_j(q) \exp(-B_j q^2) \exp[2\pi i(hx_j + ky_j + lz_j)] \quad (2.3)$$

summed over the symmetry-expanded content of the unit cell, with  $n_j$  the site occupancy. Symmetry operations are taken from the CIF as written, parsed into a rotation and a translation, applied to each asymmetric-unit site, and duplicate positions removed within a tolerance of  $10^{-4}$  in fractional coordinates.

## 2.3 The one-dimensional layer model

For a basal reflection  $h = k = 0$ , equation (2.3) reduces to

$$F(00l) = \sum_j n_j f_j T_j \exp(2\pi i l z_j) \quad (2.4)$$

which depends only on the fractional coordinate along  $c$ , whatever the cell angles: the scalar product of the reciprocal vector  $(00l)$  with a position  $x\mathbf{a} + y\mathbf{b} + z\mathbf{c}$  is  $lz$  exactly. No projection correction is required. The basal spacing follows from the triclinic metric as

$$d(001) = 1/|\mathbf{c}^*| = V / (ab \sin\gamma) \quad (2.5)$$

A *layer model* is therefore a list of scatterers at absolute heights  $z$  (in ångström) within one layer of repeat distance  $d$ , each with an occupancy and a displacement parameter, together with the continuous layer structure factor

$$F(s) = \sum_j n_j f_j(s/2) \exp(-B_j s^2/4) \exp(2\pi i s z_j) \quad (2.6)$$

Here  $s$  is continuous, not restricted to  $l/d$ : an interstratified stack is not periodic and scatters at every  $s$ , which is precisely what §2.5 integrates.

## Deriving a layer model from a polytype cell

The structures available for illite, chlorite and kaolinite are multi-layer polytypes: illite ICSD 90144 is a two-layer  $C2/c$  cell, kaolinite 2M ICSD 30285 a two-layer  $C1c1$  cell. Successive layers in these polytypes are related by operations of the form  $(x, -y, z+1/2)$  — a rotation or reflection in the  $ab$  plane combined with a translation of exactly half the cell along  $c$ . Such an operation leaves the projection of the electron density on  $c^*$  unchanged. The projected density therefore repeats every  $d(001)/n$ , and a single-layer model is obtained exactly by folding all atomic heights modulo that distance. This is done automatically; the fold is exact for the  $2M_1$  and  $1Ib$  polytypes used here and was verified numerically (§A.1).

**Consequence.** Because only the projection on  $c^*$  enters, kaolinite 1M and kaolinite 2M have nearly identical basal series and are not distinguishable from 00l intensities alone. They are separable only through their  $hkl$  reflections, which in a

strongly oriented mount are weak. Both are provided, and a fit that divides intensity between them should not be read as a polytype determination.

## 2.4 Basal reflections of a periodic stack

For a crystallite of  $N$  identical layers of thickness  $d$ , the amplitudes add with a phase  $2\pi s d$  per layer and

$$I(s) = |F(s)|^2 \cdot \sin^2(\pi N s d) / \sin^2(\pi s d) \quad (2.7)$$

the familiar Laue interference function, which has maxima of height  $N^2$  at  $s = l/d$  and a width in  $s$  of about  $1/(Nd)$ . Equation (2.7) is the  $A = B$  limit of the general result below, and is used as a closed-form check on it (§A.1).

## 2.5 Interstratification: the matrix method

Consider a crystallite of  $N$  layers stacked along  $c^*$ , each of type A or B, with layer structure factors  $F_A(s)$ ,  $F_B(s)$  and thicknesses  $d_A$ ,  $d_B$ . Let the sequence be a stationary first-order Markov chain with transition matrix  $\mathbf{P}$ , where  $P_{mn}$  is the probability that a layer of type  $m$  is followed by one of type  $n$ , and with stationary proportions  $\mathbf{W}$  satisfying  $\mathbf{W}\mathbf{P} = \mathbf{W}$ .

The scattered amplitude is

$$A(s) = \sum_{j=1}^N F_{t_j}(s) \exp(2\pi i s z_j), \quad z_j = \sum_{k < j} d_{t_k} \quad (2.8)$$

and the expected intensity  $\langle |A|^2 \rangle$  separates into a self term and a pair term,

$$I(s) = \sum_j \langle |F_{t_j}|^2 \rangle + 2 \operatorname{Re} \sum_{j < k} \langle F_{t_k} F_{t_j}^* \exp(2\pi i s(z_k - z_j)) \rangle \quad (2.9)$$

Define the matrix

$$T_{mn}(s) = P_{mn} \exp(2\pi i s d_m) \quad (2.10)$$

Then the sum over all sequences of  $n$  steps from type  $m$  to type  $p$ , weighted by probability and carrying the accumulated phase, is exactly  $[\mathbf{T}^n]_{mp}$ . Writing  $f_m = W_m F_m^*$ , the pair at separation  $n$  contributes  $\mathbf{f}^T \mathbf{T}^n \mathbf{F}$ , and counting the  $(N-n)$  pairs at that separation gives

$$I(s) = N \sum_m W_m |F_m|^2 + 2 \operatorname{Re} \sum_{n=1}^{N-1} (N-n) \mathbf{f}^T \mathbf{T}^n \mathbf{F} \quad (2.11)$$

which is the result implemented. Dividing by  $N$  and letting  $N \rightarrow \infty$  gives the Kakinoki–Komura form

$$I_\infty(s)/N = \operatorname{Re} \{ \mathbf{f}^T [2(\mathbf{I} - \mathbf{T})^{-1} - \mathbf{I}] \mathbf{F} \} \quad (2.12)$$

since  $\sum_{n \geq 1} \mathbf{T}^n = (\mathbf{I} - \mathbf{T})^{-1} - \mathbf{I}$  and  $\mathbf{f}^T \mathbf{F} = \sum W_m |F_m|^2$ . ClayQuant uses the finite form (2.11), because crystallite thickness controls basal peak width and the infinite limit would give delta functions.

**On the phase convention.** The distance from the origin of layer  $j$  to that of layer  $j+1$  is the thickness of layer  $j$ , not of layer  $j+1$ . This is why the exponential in (2.10) carries  $d_m$ , the thickness of the layer being *left*. Using the other convention transposes the roles of the two components and is a common source of error; the implementation is checked against the pure end members and against the closed form (2.7) for this reason (§A.1).

## 2.6 Crystallite thickness

Real specimens contain crystallites of many thicknesses. Averaging (2.11) over a distribution  $p(N)$  and normalising per layer gives

$$I(s)/\langle N \rangle = \sum_m W_m |F_m|^2 + 2 \operatorname{Re} \sum_{n \geq 1} G(n) \mathbf{f}^T \mathbf{T}^n \mathbf{F} \tag{2.13}$$

with the pair weights

$$G(n) = \langle \max(N-n, 0) \rangle / \langle N \rangle \tag{2.14}$$

so the thickness distribution enters through one vector of weights, computed once and reused at every  $s$ . The distribution used is lognormal in  $N$ , parameterised by its mean  $\langle N \rangle$  and by the width  $\beta$  of the underlying normal distribution in  $\ln N$ , following the crystallite size distribution (CSDS) treatment standard in this field (Drits, Eberl & Środoń).  $G(n)$  is normalised so that  $G(0) = 1$ .

Crystallite thickness is the dominant control on basal peak width. For a 10 Å layer, a mean of 10 layers gives a 001 width of order  $0.8^\circ 2\theta$ , while a well-crystallised illite may show  $0.12^\circ$ , which requires of order 70 layers (§A.7). It cannot be assumed; it must either be spanned by the library or determined from the data.

## 2.7 Ordering and Reichweite

The transition matrix carries the ordering. For a two-component system with proportions  $W_A, W_B$ :

| Case             | Transition probabilities     | Description                                                                                                              |
|------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Random, $R = 0$  | $P_{AB} = W_B, P_{BA} = W_A$ | a layer's identity is independent of its neighbour                                                                       |
| Ordered, $R = 1$ | $P_{AB} > W_B$               | unlike layers tend to alternate; the limit $P_{AB} = 1$ at $W_A = \frac{1}{2}$ is a regular alternation (rectorite-like) |
| Segregated       | $P_{AB} < W_B$               | like layers cluster; the limit $P_{AB} = 0$ is a physical mixture                                                        |

Ordering is expressed through a single parameter  $P_{AB}$ , from which the remaining probabilities follow by the stationarity and normalisation conditions  $W_A P_{AB} = W_B P_{BA}$  and  $P_{mA} + P_{mB} = 1$ . Values of  $P_{AB}$  for which any probability would fall outside  $[0,1]$  are rejected. The supplied reference library is computed for the random case, which is the appropriate default for illite/smectite and chlorite/smectite in most sedimentary settings; ordered stacks can be computed in the simulator or by calling the library builder directly.

## 2.8 The Lorentz-polarization factor

Which Lorentz factor applies to an oriented clay aggregate measured on a diffractometer is a genuine and long-standing question, and the choice matters: the two candidate conventions differ by a factor  $l$  across a basal series after normalisation at 001, so an incorrect choice distorts every relative basal intensity and therefore every quantification derived from them.

The two candidates are the random-powder (Debye–Scherrer) form

$$LP_{\text{powder}} = (1 + \cos^2 2\theta) / (2 \sin^2 \theta \cos \theta) \quad (2.15)$$

and the single-crystal form appropriate to a one-dimensionally periodic, perfectly oriented specimen,

$$LP_{\text{crystal}} = (1 + \cos^2 2\theta) / (2 \sin 2\theta) \quad (2.16)$$

Bradley (1954) and Schoen (1962) found the random-powder factor to work for oriented aggregates; MacEwan, Amil & Brown (1961) argued for something intermediate. The matter was settled empirically by Reynolds (1965), who showed for an ethylene glycol–montmorillonite complex (his Table 2) and independently for a dry Na-montmorillonite (his Table 4) that the random-powder factor is *required* to reproduce observed basal intensities, the single-crystal factor giving large deviations — for the dry montmorillonite, a calculated 003 intensity of 141 against an observed 55.

**ClayQuant therefore uses (2.15) by default.** This is not merely a convention: the published one-dimensional clay layer models, including the glycol–smectite model used here, were refined under it, and their structure factors are only consistent when it is applied. Equation (2.16) is selectable for comparison. A monochromator-modified polarization factor,

$$p = (1 + \cos^2 2\theta \cos^2 2\theta_M) / (1 + \cos^2 2\theta_M) \quad (2.17)$$

is available for a crystal-monochromated beam.

## 2.9 Preferred orientation

Texture is described by the March–Dollase model (March 1932; Dollase 1986). For a reflection whose scattering vector makes an angle  $\alpha$  with the texture axis,

$$P(\alpha; r) = (r^2 \cos^2 \alpha + \sin^2 \alpha / r)^{-3/2} \quad (2.18)$$

The texture axis is  $c^*$ , the platelet normal, which in Bragg–Brentano reflection geometry coincides with the specimen normal.  $r = 1$  is a random powder;  $r < 1$  describes platelets lying in the specimen plane, enhancing 00 $l$  and suppressing  $hk0$ . The function conserves total scattered intensity, its average over the sphere being unity, so  $r$  redistributes intensity rather than creating it.

**Preferred orientation does not change a purely basal pattern.** For every 00 $l$  reflection  $\alpha = 0$ , so  $P = r^{-3}$  for all of them alike: a common scale factor. The March–Dollase parameter therefore has no effect on the *shape* of a basal series, and acts only on the ratio of basal to non-basal intensity. This is why the ten orientation parameters of the reference library change an interstratified pattern only through its  $hkl$  contribution (§2.12), and why  $r$  cannot be determined from basal reflections alone.

## 2.10 Beam overflow

The Lorentz factor diverges as  $1/\sin^2\theta$ , but a real measurement does not, because at low angle the irradiated length of a flat specimen,

$$L_{\text{irr}}(\theta) = R \gamma / \sin\theta \quad (2.19)$$

with  $R$  the goniometer radius and  $\gamma$  the equatorial divergence in radians, exceeds the specimen and part of the beam misses it. Only the intercepted fraction diffracts:

$$\varphi(\theta) = \min[1, L_{\text{spec}} \sin\theta / (R \gamma)] \quad (2.20)$$

This matters precisely where clay basal reflections lie. Reynolds (1965) avoided the correction by changing beam slits with angle so that the beam never exceeded his specimen, following Klug & Alexander (1954); with the fixed slits of a modern instrument it must be applied, or every calculated pattern carries an unphysical low-angle ramp. The specimen length, goniometer radius and divergence are settable; for a variable-divergence (automatic) slit, which keeps the irradiated length constant, the correction should be disabled.

## 2.11 Emission profile and peak shape

The source is described as a sum of Lorentzian-weighted lines, each with its own wavelength, relative intensity and intrinsic width. The default Cu K $\alpha$  profile is the five-line description

| Line | $\lambda$ (Å) | Relative weight | Width  |
|------|---------------|-----------------|--------|
| 1    | 1.534753      | 0.0159          | 3.6854 |
| 2    | 1.540596      | 0.5691          | 0.4370 |
| 3    | 1.541058      | 0.0762          | 0.6000 |
| 4    | 1.544410      | 0.2517          | 0.5200 |
| 5    | 1.544721      | 0.0871          | 0.6200 |

Every reflection is generated once per emission line, at the  $2\theta$  corresponding to that line's wavelength and scaled by its weight, so the K $\alpha_1$ /K $\alpha_2$  separation emerges naturally and correctly with angle rather than being imposed. Other profiles, including a single K $\alpha_1$  line, are available.

The instrumental profile is a pseudo-Voigt whose Gaussian width follows the Caglioti relation

$$\text{FWHM}^2 = U \tan^2\theta + V \tan\theta + W \quad (2.21)$$

with a mixing parameter  $\eta$  between Gaussian and Lorentzian, optionally combined with a Scherrer broadening from a finite in-plane crystallite dimension. Note that broadening along  $c^*$  is *not* imposed through (2.21): for interstratified phases it arises physically from the crystallite thickness distribution of §2.6.

## 2.12 Assembling a pattern

A pattern of a discrete crystalline phase is computed by enumerating every reciprocal lattice point within the measurable range, evaluating (2.3), and applying preferred orientation, Lorentz–polarization and beam overflow before convolution with the emission profile and peak shape. Reflections are enumerated individually rather than by symmetry-equivalent family, so multiplicity is accounted for by the enumeration itself and each reflection receives its own March–Dollase factor according to its own angle to  $c^*$  — which is what the texture correction requires.

An interstratified pattern is assembled from two contributions:

1. the continuous basal intensity of §2.5, evaluated at every point of the  $2\theta$  grid and carrying LP, beam overflow and the emission profile;
2. the non-basal ( $hkl$ ,  $k \neq 0$  or  $h \neq 0$ ) reflections of the host phase, computed from its crystal structure and scaled by their March–Dollase factors.

**Approximation.** Contribution (2) treats the non-basal scattering of an interstratified crystal as that of its host layer type, scaled by the host's proportion. This is an approximation. In reality, turbostratic stacking produces two-dimensional  $hk$  bands with a characteristic sawtooth profile rather than sharp three-dimensional reflections, and the interstratified component contributes its own non-basal scattering. The approximation is acceptable for a strongly oriented mount, where non-basal intensity is heavily suppressed by the texture, and it degrades as  $r$  approaches 1. The relative scaling of the two contributions is fixed by computing, for a pure host stack, the ratio of its basal intensity in the two formulations, so that a pure end member reproduces its own three-dimensional calculation (§A.1).

## 2.13 Quantification

The measured pattern  $y(2\theta)$ , after zero-error correction and background subtraction, is fitted as

$$y(2\theta) \approx \sum_p c_p y_p(2\theta), \quad c_p \geq 0 \quad (2.22)$$

over the library patterns  $y_p$ , by non-negative least squares. The non-negativity constraint is physical — a phase cannot be present in negative amount — and has the useful side effect of producing a sparse solution, so that of several hundred candidate patterns only a handful take non-zero coefficients.

The solve is weighted. The variance of a counting measurement at a point is the number of counts recorded there, so the weight is

$$w_i = 1 / \max(y_{\text{raw},i}, 1) \quad (2.23)$$

taken from the *raw* counts, before background subtraction: subtracting a background does not reduce the Poisson variance of the measurement. Weighting instead by the background-subtracted intensity — an easy error — places the greatest weight on the flat regions between peaks, where the subtracted value is near zero and only noise remains, and almost none on the peaks that carry the information (§A.9). Agreement is reported as

$$R_{\text{wp}} = [ \sum w_i (y_i - y_{\text{calc},i})^2 / \sum w_i y_i^2 ]^{1/2}, \quad R_p = \sum |y_i - y_{\text{calc},i}| / \sum |y_i| \quad (2.24)$$

## From coefficients to proportions

Two measures are reported for each phase, and they answer different questions:

### Scattering share

the phase's fraction of the total calculated integrated intensity,  $c_p \int y_p / \sum c_q \int y_q$ . This is the quantity the diffractogram actually measures.

### Amplitude share

the phase's fraction of the summed scale factors,  $c_p / \sum c_q$ . Broad interstratified patterns carry far more integrated area per unit peak height than a sharp discrete phase, so the two can differ by a factor of several.

## A parameter the library stopped rather than determined

The fit chooses among pre-computed patterns (§3.3), so it cannot return a value the library does not hold. When a phase's coefficients collect on the lowest or highest value of a spanned parameter, the honest reading is not that the parameter equals that value but that the data wanted to go further and ran out of library: the answer is a bound, not a measurement. In a table of numbers that all look alike that difference is invisible, and it is the same signal that marks a refined parameter sitting on its limit — worth distrusting wherever it appears.

So every fit reports it. `parameters_at_an_edge` lists each phase and parameter whose fitted weight piles up on an end of its range, and the fit records the list in its metadata without being asked. Merely including an end value among others is not flagged; the default threshold is 80 % of a phase's coefficient.

Two cases look identical in a table of numbers and are not the same thing, so they are worded differently. A crystallite thickness sitting on the largest value calculated means the data wanted a larger one and the library should be extended. A composition sitting on 1.00 means the data wants the pure phase, and there is nothing beyond it to extend to — the fit has answered rather than run out of room, and the same goes for an orientation parameter at  $r = 1$ , which is a random powder. Advice to span those further is advice that cannot be taken, and a warning a user learns to ignore is worse than none.

What prompted it: a measured pure illite standard put 94 % of its illite on the 0.99 entry of the illite/smectite series, the highest the series went. That reported 1 % of smectite the specimen did not have, and reported the end of a range as though it were a composition. The series now reaches 1.00 (§3.3), and the chlorite/smectite series with it.

## How much of an interstratified phase is the host

The library spans illite/smectite from 0.20 to 0.99 illite layers and groups every one of those entries under the phase name I/S. The name alone is therefore misleading at the ends of the series: a fit that settles on the 0.99/0.01 entry has found a stack of essentially pure illite, and a table reporting that as “I/S, 63 % of the pattern” reads as though 63 % of the specimen were an expandable clay. On a measurement of a pure illite standard that is the difference between accepting the material and rejecting it.

So each phase also reports the host-layer fraction the fit chose, averaged over its entries and weighted by their coefficients, with the expandable fraction as its complement. A discrete mineral reports 1. Read the two together: I/S at 0.99 is an illite, at 0.60 it is genuinely interstratified.

## Weight percent

A third quantity is reported, and it is a weight percent. It needs no reference intensity ratios, because the calculated patterns carry the mass of what they were calculated from. A stored pattern is the calculated one divided by a normalisation  $s_c$ ; the calculated one

is for a single unit of the phase — one unit cell for a discrete mineral, one layer for an interstratified stack — of mass  $M_e$ . So a fitted coefficient  $c_e$  counts units, and

$$W_p \propto k_p \sum_{e \in p} c_e M_e / s_e \quad (2.25)$$

with the constant of proportionality the same for every phase — it is the instrument, the irradiated volume and the counting time, which the phases of one measurement share — so the ratios are ratios of mass. This is the relation Rietveld analysis writes as  $W \propto S(ZMV)$ , in the form ClayQuant's normalisation calls for. Tested by building a specimen from known masses of library patterns and asking for the masses back, it returns them exactly (§A.14).

**$k_p$  is 1, and that is the whole difficulty.** Equation (2.25) assumes the calculated pattern accounts for everything the phase contributes. In an oriented mount it does not, because texture differs between minerals and between preparations, and the fit cannot tell a well-oriented phase from an abundant one. The effect is not a correction of a few percent. A basal series is multiplied by  $r^{-3}$  (§2.9), so a clay fitted at  $r = 0.13$  is calculated to scatter 455 times more per gram than the same clay at  $r = 1$ , and the mass inferred for it falls by the same factor, while quartz — which does not plate onto the slide and is fitted at  $r = 1$  — takes the mass the clays lose. **Until  $k_p$  has been measured for a preparation, read the weight percent as indicative and the clay-to-non-clay split as the least reliable part of it.** The fitted  $r$  is reported beside every phase so this is visible rather than hidden.

**Measuring  $k_p$ .** It is one factor per phase, covering texture, microabsorption and any error in the structural model together. To determine it, analyse specimens whose composition is known independently, through the same preparation routine, and solve for the factors that reproduce that composition — `Calibration.from_known_composition` does the arithmetic, and the factors are stored in a JSON file that the analysis loads. One way to obtain the known composition is to measure the same separate twice: once as a random mount spiked with an internal standard, which gives absolute weight fractions by the usual internal-standard method and is unaffected by texture because there is none; and once as the oriented mounts. The random mount fixes the composition, the oriented fit then yields  $k_p$ . An oriented mount cannot be spiked usefully, since the standard does not orient as the clays do and its share in the mount is not its share in the suspension.

The other route needs no mixture at all: mount one pure mineral, and measure a standard of known weight separately through the same optics. Then the mass the specimen holds is known independently of any fit, and  $k_p$  is what stands between it and the mass (2.25) infers. That is the external-standard method below, and §A.18 reports it.

The older route is still available for a laboratory that has measured reference intensity ratios of its own: `Quantification.weight_percent` divides each phase's share of diffracted intensity by its ratio.

## An internal standard, and what the fit does not account for

Every weight percent above is a share of what was *fitted*, and sums to 100 however much of the specimen was left out. A mount holding amorphous material, or a mineral the library does not have, therefore reports every phase in it overstated by one common factor, and nothing inside the fit reveals the factor: the residual of a whole-pattern fit is not a measure of missing crystalline matter, because a missing phase's intensity is taken up by whatever else lies under it.

A weighed addition of a standard reveals it. If a mass fraction  $W_{\text{std}}$  of a standard is mixed into the specimen before mounting, and the fit returns  $W'_{\text{std}}$  for it on the relative basis, then every phase's weight percent *of the specimen* is

$$W_p = W'_p \cdot W_{\text{std}} / W'_{\text{std}} \quad (2.26)$$

and what is left over,  $100 - \sum_p W_p$ , is the amorphous content together with any crystalline phase absent from the fit. The measurement separates that sum from the phases that were fitted, but not into its two parts.

Pass the standard and its weight to the analysis (`quantify(..., internal_standard=("Corundum", 20.0))`) and the table gains an absolute column beside the relative one, with the unaccounted fraction reported alongside. A standard that was weighed in but does not appear in the fit leaves the analysis relative and says so, rather than scaling by a number it does not have. The absolute column is given on the whole-specimen basis only: a clay-mineral table is renormalised to 100 % by definition, so scaling it would mean nothing.

**An internal standard belongs in a random mount, not an oriented one.** The standard must share the specimen's texture for (2.26) to hold, and a spike added to a clay suspension does not: it does not plate onto the slide as the clays do, so its share of the mount is not its share of the suspension. Equation (2.26) is for a pressed random powder. This is also why the internal-standard route cannot by itself measure  $k_p$  for an oriented mount (§A.15); an *external* standard can, and is treated next.

## An external standard, and the absolute scale

The restriction just stated — that a standard mixed into an oriented mount cannot work — does not apply to a standard that is never mixed in. Measure a standard of known weight fraction on its own, under the same optics, and it is never asked to share the specimen's texture. That is the external-standard method of O'Connor and Raven (1988), and it is what turns the proportionality of (2.25) into an equality.

The standard's own measurement fixes one instrument constant,

$$K = S_{\text{std}}(ZMV)_{\text{std}} \cdot (\mu/\rho)_{\text{std}} / W_{\text{std}} \quad (2.27)$$

and any later measurement on the same instrument then gives an absolute weight fraction without a spike,

$$W_p = S_p(ZMV)_p \cdot (\mu/\rho)_{\text{spec}} / K \quad (2.28)$$

where  $S(ZMV)$  is, in ClayQuant's normalisation, exactly the sum  $\sum c_e M_e / s_e$  of (2.25), and  $\mu/\rho$  is the mass attenuation coefficient of the whole specimen at the wavelength used. Absorption enters because the two measurements sample different depths: a weakly absorbing specimen returns intensity from further down, and the ratio of the two coefficients corrects for it. `absorption.py` computes  $\mu/\rho$  for a structure from its mass fractions and tabulated elemental coefficients at Cu K $\alpha$ , and refuses — rather than guessing — for an element it does not hold.

Two conditions have to hold, and both are checkable rather than assumed. The specimen must be thicker than the beam penetrates, since (2.28) is derived for infinite thickness; `penetration_depth` and `thick_enough` give the depth  $\sin\theta/2\mu$  and test a film against it, which for a clay mount matters most at low angle where the depth is smallest. And the two measurements must be made through the same optics: a difference in divergence slit, mask, counting time or tube setting changes  $K$  and is not recoverable afterwards. So

ExternalStandard records the conditions of the standard's measurement and comparable\_with names every difference it finds against a later one, rather than scaling by a constant that no longer applies.

For a mount of a *known* phase this reads backwards and measures  $k_p$  directly:  $W_p$  is known, so calibration\_factor returns the ratio of the mass (2.25) infers to the mass the specimen holds. That is the one route to  $k_p$  for an oriented mount that does not need an independent analysis of a mixture, and §A.18 carries it out on four pure clay mounts against a corundum measured through the same optics.

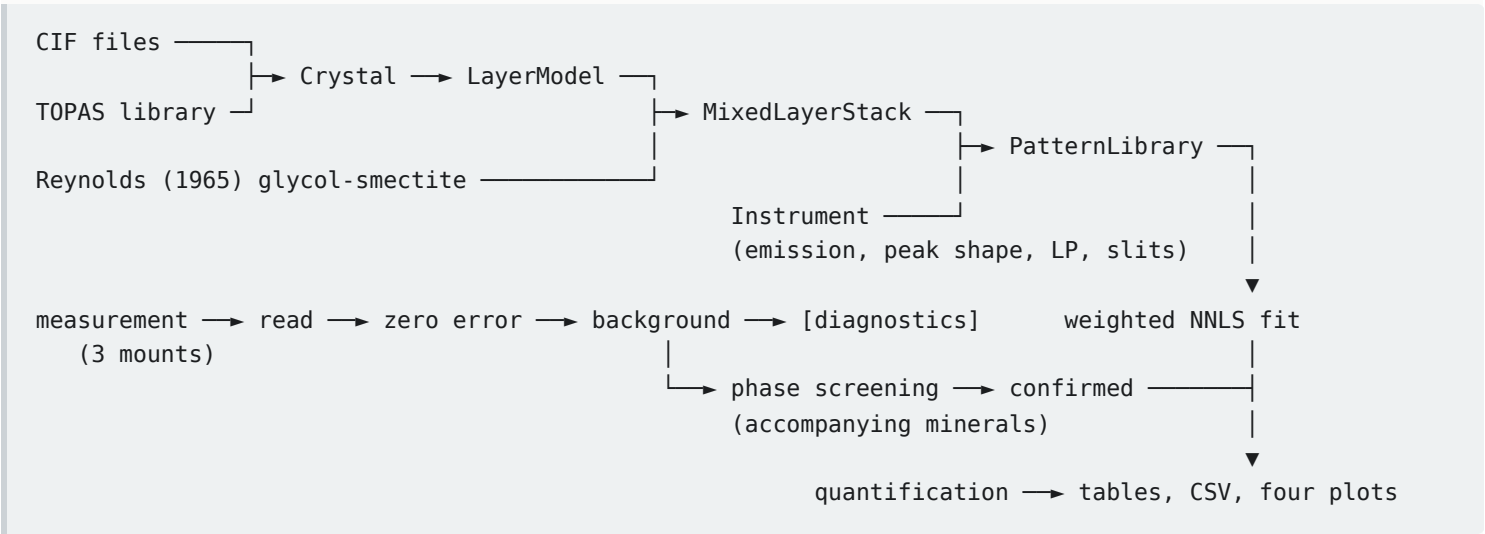
Results are given on two bases: over all fitted phases, and over the clay minerals alone, renormalised to 100 % with the accompanying minerals removed. The latter is the customary basis for reporting a clay mineral assemblage.

# 3 Implementation

## 3.1 Module map

| Module                     | Contents                                                                                                                                 |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| crystal.py                 | CIF reading, symmetry expansion, the Crystal and LayerModel classes, structure factors, derivation of a layer model from a polytype cell |
| scattering.py              | Waasmaier–Kirkel scattering factors, species normalisation, displacement factors                                                         |
| absorption.py              | mass attenuation coefficients at Cu K $\alpha$ , penetration depth, the infinite-thickness test                                          |
| emission.py                | emission profiles, including the five-line Cu K $\alpha$ description                                                                     |
| optics.py                  | Lorentz–polarization, March–Dollase, beam overflow                                                                                       |
| profile.py                 | pseudo-Voigt peak shapes, Caglioti widths, Scherrer broadening                                                                           |
| mixed_layer.py             | the matrix method, transition matrices, the CSDS and its pair weights                                                                    |
| pattern.py                 | Instrument, reflection enumeration, powder, basal and mixed-layer pattern assembly, peak lists                                           |
| library.py                 | the PatternLibrary, its builder, storage, and the spanning of orientation, thickness and spacing                                         |
| models.py                  | the expected CIF sources, the structure search path, the Reynolds (1965) glycol–smectite layer                                           |
| bern.py                    | import of a TOPAS/jEdit structure library into a phase database; clay/non-clay classification                                            |
| io.py                      | readers for XRDML, .xy, Bruker .raw and two-column text                                                                                  |
| pairing.py                 | recognising the three mounts of one sample from their file names                                                                         |
| calibration.py             | 2 $\theta$ zero error on the quartz 100 reflection, merit profile for the slider                                                         |
| background.py              | background components, peak stripping, the non-parametric background                                                                     |
| diagnostics.py             | kaolinite collapse and expandable-clay swelling                                                                                          |
| detection.py               | finding accompanying minerals, by competitive fit or by peak matching                                                                    |
| npls.py                    | the weighted non-negative least-squares fit and its metrics                                                                              |
| quantification.py          | grouping by phase, the two normalisation bases, CSV export                                                                               |
| plots.py                   | the four result figures and the report writer                                                                                            |
| gui/                       | the browser interface and its session state                                                                                              |
| install.sh,<br>install.ps1 | installers for Linux/macOS/WSL and for Windows: interpreter selection, environment creation, installation and verification               |

### 3.2 Data flow



### 3.3 The reference library

Because the fit selects among pre-computed patterns rather than refining parameters, the library must span the parameters that actually vary between specimens. Three are spanned:

| Dimension                           | Values                                                  | Why it must be spanned                                                                                                          |
|-------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Preferred orientation <i>r</i>      | 0.1 to 1.0 in steps of 0.1                              | texture is a property of the preparation, not of the mineral; it sets the ratio of basal to non-basal intensity                 |
| Mean crystallite thickness <i>N</i> | 5, 15, 50 layers                                        | sets basal peak width; 10 layers give a 001 width near 0.8° where a well-crystallised illite shows 0.12°                        |
| Layer spacing <i>d</i>              | illite 9.95, 10.02, 10.10 Å;<br>chlorite 14.20, 14.35 Å | illite <i>d</i> (001) varies with interlayer potassium and hydration; an error of 0.07 Å displaces the 001 by half a peak width |

Together with the compositional series this gives, with default settings, 1751 patterns:

| Phase                             | Patterns | Composition range                          |
|-----------------------------------|----------|--------------------------------------------|
| Illite/smectite (I/S)             | 1440     | 16 compositions, 0.20/0.80 to 1.00/0.00    |
| Chlorite/smectite (C/S)           | 240      | 1.00/0.00, 0.95/0.05, 0.90/0.10, 0.85/0.15 |
| Illite                            | 30       | discrete                                   |
| Chlorite                          | 20       | discrete                                   |
| Kaolinite 1 <i>M</i> , 2 <i>M</i> | 10 + 10  | discrete                                   |
| Glycol-smectite                   | 1        | pure end member                            |

**Both series reach their pure end member**, and that matters more than it looks. Without it the closest a pure illite could be described as was 0.99/0.01, and a fit of one piles its whole coefficient onto that entry: it reports 1 % of smectite the specimen does not have, and it reports the end of a range as though it were a composition (§2.13). A measured pure illite

standard is what showed this, putting 94 % of its illite on the 0.99 entry. The end member is also not redundant with the discrete illite entries: those are three-dimensional powder patterns whose basal widths come from the instrumental profile, while the interstratified entries carry the crystallite thickness distribution of §2.6, which is what actually sets a basal width. Only through this series can a pure illite be fitted with its thickness spanned.

Each pattern is stored normalised to unit maximum *above*  $4^\circ 2\theta$ , not to its global maximum. Intensity rises steeply towards zero angle — the Lorentz factor still diverges after the beam-overflow correction — and no clay basal reflection of interest lies below about  $4^\circ$ ; normalising on that rise would shrink the diffraction peaks themselves and leave the fit poorly conditioned.

Each pattern also has its own smooth continuum stripped before storage. An interstratified stack of finite thickness scatters a genuine diffuse continuum rising towards zero angle, but in a measurement that continuum is inseparable from air scatter and the direct beam and is always treated as background. Leaving it in the basis patterns would let a phase absorb the measured background and inflate its apparent share.

For that reason every pattern is *calculated* over a wider range than it is stored on: one stripping window beyond each end, trimmed away afterwards. Peak stripping needs data on both sides of a point, so its estimate is unreliable within one window of either end (§A.11), and for a calculated clay pattern that is exactly where the diffuse continuum is largest. Moving the ends outside the stored range takes the continuum left standing at  $3.5^\circ$  on an I/S 0.80/0.20 pattern from 21 % of its strongest peak down to 5 %, with the peaks themselves unchanged. The lower limit of the calculation is  $1^\circ 2\theta$ , below which the Lorentz factor is of no use.

### 3.4 Numerical notes

- Symmetry operations are parsed by an explicit tokeniser, never by evaluating file content as code.
- The matrix sum (2.13) is truncated at the largest  $N$  carrying appreciable weight; the  $2 \times 2$  matrix powers are accumulated once for all grid points simultaneously.
- Reflection enumeration is bounded by the measurable  $d_{\min}$  at the longest wavelength in the emission profile.
- The NNLS solve is that of Lawson & Hanson as implemented in SciPy, applied to the row-weighted system; with 1600 columns and about 1800 points a fit takes well under a second.

## 4 Installation and structure data

### 4.1 Installation

#### Requirements

| Component        | Requirement   | Notes                                                                     |
|------------------|---------------|---------------------------------------------------------------------------|
| Python           | 3.10 or later | developed and tested on 3.11                                              |
| NumPy            | $\geq 1.24$   | required                                                                  |
| SciPy            | $\geq 1.10$   | required, for the NNLS solver                                             |
| Dash             | $\geq 2.14$   | extra gui; needed only for the interface                                  |
| Plotly           | $\geq 5.18$   | extra gui; also used by the figure export                                 |
| gemmi            | $\geq 0.6$    | extra import; needed only to import a TOPAS structure library             |
| pytest           | $\geq 7.4$    | extra dev; needed only to run the test suite                              |
| Operating system | any           | pure Python; no compiler required. Linux, macOS and Windows               |
| Disk             | ~200 MB       | dependencies, plus 30–60 MB per stored reference library                  |
| Memory           | ~1 GB         | a default library holds 1601 patterns on the measurement grid             |
| Browser          | any current   | the interface runs locally at 127.0.0.1; nothing is sent over the network |

#### Procedure

The supplied installer does the whole of this and checks the result. From the project directory:

```
# Linux, macOS, WSL
./install.sh

# Windows PowerShell
.\install.ps1
```

It selects an interpreter, builds the virtual environment, installs ClayQuant with its dependencies, verifies that the package imports, checks that every command it declares has been created and answers, and reports which clay structures it can find. Re-running it is safe: an environment of the right version is reused, one of the wrong version is rebuilt. Useful options are `-y` to answer prompts automatically, `--python /path/to/python3.12` to force an interpreter, and `--no-dev` to omit the test and import extras.

#### Starting it from an icon

At the end of the installation you are asked whether to create a desktop icon and a menu entry. Answering yes writes what the desktop in question expects — a `.desktop` file on Linux, an application bundle in `~/Applications` on macOS, a shortcut on the desktop and in the Start menu on Windows — pointing at the interpreter of the environment ClayQuant was installed in, so nothing has to be activated or on PATH. Double-clicking it starts the program and opens it in the default browser.

It can be done or undone at any time, and does not need the installer:

```
./clayquant shortcut      # create them
./clayquant shortcut --where  # say where they would go
./clayquant shortcut --remove # take them away
```

For the icon, put an image called `clayquant` into `assets/` and run the command again. Any common raster format will do and the spelling of the name does not matter — `ClayQuant.PNG` is found as readily as `clayquant.png`, which on a case-sensitive filesystem it otherwise would not be — and it is converted to the format each platform needs (`.ico` on Windows, `.icns` on macOS). The placeholder that ships with ClayQuant is called `clayquant-placeholder.png` so that a supplied icon always takes precedence over it and is never overwritten by an update. The command prints which file it used.

**Run it again after every update.** The commands (`clayquant-gui` and the rest) are written into the environment when the package is installed and at no other time, so an environment built from one version keeps that version's set of commands. Updating the source with `git pull` does not add a command the new version introduced, and it is then reported as not found although its code is in the checkout. Re-running the installer writes them all; §7 gives the module form to use meanwhile.

**Why an installer rather than three commands.** `python -m venv` runs whichever interpreter the name `python` resolves to, and *each Python version carries its own venv module and ensurepip*, with its own bundled wheels. On Ubuntu and WSL, `python` and `python3` are frequently an older version than the one that was installed later, so the command fails with

```
The virtual environment was not created successfully because ensurepip is
not available. ... apt install python3.10-venv
```

even on a machine where `python3.12-venv` is installed. Nothing is failing to "see" the newer Python: version 3.10 was invoked, and 3.10's package is the one missing. The immediate fix is to name the interpreter — `python3.12 -m venv .venv` — and the installer does this automatically, trying every interpreter it can find, newest first, testing each by actually building a throwaway environment, and naming the exact distribution package if none succeeds.

## Doing it by hand

1. Obtain the source and create an isolated environment. Name the interpreter explicitly rather than relying on `python`.

```
git clone https://github.com/lomalaspina/ClayQuant.git
cd ClayQuant
python3.12 -m venv .venv      # or whichever version 3.10+ you have
```

2. Activate the environment.

```
# Linux and macOS
source .venv/bin/activate

# Windows, PowerShell
.venv\Scripts\Activate.ps1

# Windows, cmd.exe
.venv\Scripts\activate.bat
```

3. Install. The extras select optional components; installing all of them is normal for interactive use.

```
pip install -e ".[gui,import,dev]"
```

The editable install (-e) means edits to the source take effect without reinstalling. To install only what is needed to compute patterns from a script, omit the extras: `pip install -e .`

4. Verify the installation. The test suite runs in a few seconds.

```
pytest -q
```

Tests that need structure data are skipped automatically when it is absent, and report as s rather than as failures; see §4.2. With no structure data installed, expect 29 passed and 10 skipped; with the four CIFs in place, 45 passed. Any *failure* indicates a problem and should be reported.

5. Provide structure data (§4.2), build a reference library (§4.4), and start the interface:

```
clayquant-gui
```

which prints a local address, by default `http://127.0.0.1:8050`, to open in a browser. Use `--port` to choose another port and `--host` to bind an interface other than the loopback.

**Upgrading.** After pulling a new revision, re-run `pip install -e ".[gui,import,dev]"` to pick up any dependency change, and rebuild the reference library if the pattern calculation has changed. Stored libraries carry the metadata of the settings they were built with, and a library built by an older revision is read but may not correspond to the current calculation.

## 4.2 Structure data and licensing

**No structure data is distributed with ClayQuant.** The crystal structures it uses are ICSD entries, which are licensed by FIZ Karlsruhe and may not be redistributed. They must be exported from your own subscription. The repository's ignore rules are written so that a local copy cannot be committed by accident.

Export each of the following as CIF and put it in the structure directory. The name given is the one ClayQuant looks for first, but it is not required: a file is also accepted if its name carries the ICSD code, or the phase name with its polytype.

Kaolinite\_1M\_63192.cif and Illite\_ICSD\_90144.cif are both found, in any letter case.

| Phase                        | ICSD   | File name                   | Structure reference                           |
|------------------------------|--------|-----------------------------|-----------------------------------------------|
| Illite                       | 90144  | illite_ICSD_90144.cif       | Gualtieri (2000)                              |
| Chlorite (clinochlore IIb-4) | 164234 | chlorite_ICSD_164234.cif    | Zanazzi, Comodi, Nazzareni & Andreozzi (2009) |
| Kaolinite 1M                 | 63192  | kaolinite_1M_ICSD_63192.cif | Bish & Von Dreele (1989)                      |
| Kaolinite 2M                 | 30285  | kaolinite_2M_ICSD_30285.cif | Gruner (1932)                                 |

These directories are searched, in this order, and all of them are searched rather than only the first that exists — the program is as likely to be started from the home directory as from the project, and a file that is plainly there should not be reported missing over where the shell happened to be:

1. the path in the environment variable CLAYQUANT\_STRUCTURE\_DIR, if set;
2. a structures folder in the current working directory;
3. the current working directory itself;
4. the structures folder of a source checkout;
5. the package data directory.

To see what was looked for, where, and what was found:

```
python -c "from clayquant.models import describe_structure_search as d; print(d())"
```

The same report is printed when a structure cannot be loaded, so an error names the directories that were searched rather than one expected file name.

The glycol–smectite layer requires no file. It is the published one-dimensional model of Reynolds (1965), transcribed into the source with its provenance, and is reproduced under §A.2.

## 4.3 Importing a TOPAS structure library

The accompanying non-clay minerals — quartz, feldspars, carbonates, sulphates, oxides — come from a structure library of your own. ClayQuant reads the jEdit macro-menu form in which TOPAS structure collections are commonly distributed: an XML file whose items each insert a TOPAS str block with a phase name, cell parameters, space group and site lines.

```
clayquant-import-structures my_structures.xml -o structures/phases.json
```

Space group symbols are expanded to explicit symmetry operations with gemmi, including the origin-choice settings that TOPAS writes as a trailing letter (Fd-3mS for origin choice 1, Fd-3mZ for choice 2; see §A.3). The operations are stored in the generated database, so gemmi is not needed afterwards.

The generated database is derived from licensed structure data and is likewise kept out of version control.

## Structures from one of your own refinements

TOPAS writes a refinement's result back in the syntax it read, so a refinement's `.out` — or its `.inp` — is itself a structure library, of the few phases that refinement contained. The same command reads it, and decides which kind of file it has by looking:

```
clayquant-import-structures M_26_1041_air.out -o structures/refined.json
```

Several sources may be given, in increasing order of authority: where two name the same phase the later one is kept, so a refinement laid over a library keeps the library's breadth and takes the refinement's structure for the phases it covers.

```
clayquant-import-structures my_structures.xml M_26_1041_air.out -o structures/phases.json
```

For the clay minerals the refined structure goes further than the phase database: it can replace the published CIF the pattern library is calculated from.

```
clayquant-build-library --refined-structures M_26_1041_air.out
```

Which phases to take is chosen per mineral, with `--refined-only` or `--refined-except`, because the judgement differs between them. Keeping a published structure that can be cited, while taking the refined ones that describe the specimen better, is a reasonable decision rather than a missed opportunity:

```
clayquant-build-library --refined-structures M_26_1041_air.out --refined-except chlorite
```

This is worth doing, and worth doing for the clays in particular, because the octahedral Fe-for-Mg and Fe-for-Al substitutions are refinable against the basal intensities, differ from one deposit to the next, and change those intensities substantially — iron scatters about twice as strongly as the magnesium it replaces. On the specimen of §A.16 it improves  $R_{wp}$  from 35.9 % to 31.2 %, and the occupancies rather than the refined cells account for nearly all of that.

**Do not extend it to the accompanying minerals without checking.** Measured on the same specimen, taking their refined cells as well *costs* 3 points of  $R_{wp}$  (§A.16). Part of the reason is identifiable — quartz is also the zero-error standard (§5.3), so replacing its structure puts the calibration and the fitted cell out of step — and part is not. Where a library structure is *wrong* rather than merely someone else's, overriding it is still right: a refinement is how the incomplete cordierite of §A.14 was found to be incomplete, and supplies the sites it is missing.

A refinement that models its clays as `hkl_Is` peak sets, which is the usual way to refine an oriented mount, contains no clay structure at all; the import says so rather than reporting success with nothing in it. A refinement also covers only the phases it contained: one that fits kaolinite as the *2M* polytype supplies no *1M*, and the CIF for that is still wanted. The refined structures replace the CIFs one phase at a time; they do not remove the need for the folder.

## Or as a CIF in the structure folder

The refined structures do not pass through the structure folder at all — the CIF files there are untouched, and the substitution happens when the pattern library is calculated. There is a second route, though, and it is the one most people will reach for first: put

a CIF carrying the refined values into the folder like any other. It works, and TOPAS will have written the file already if the refinement carried `Out_CIF_STR( . . . )` for the phase.

Two details of such a file are handled, because both would otherwise cost the user the change they were trying to make. TOPAS writes a mineral name as `?Clinochlore?`, using `?` where CIF wants a quote, and the marks are stripped rather than kept as part of the name. And the file arrives named after the phase as the refinement called it, so a chlorite refined as *Clinochlore* arrives as `Clinochlore.cif`, which carries neither the ICSD code nor the key `chlorite`; the mineral's other names are now consulted, where before such a file matched nothing, the published structure was used instead, and nothing said so.

**Which route to use is not a matter of taste.** The structure folder is global: one file serves every specimen analysed afterwards. A refined occupancy belongs to *one* specimen — the iron content of a chlorite is a property of the deposit, and two specimens from different formations refine to octahedral compositions that are not close to each other. Put one specimen's refined chlorite in the folder and every later sample is calculated with that specimen's iron content, which is a harder mistake to notice than a missing file. `--refined-structures` keeps the coupling visible and per-analysis: the library is built from the refinement of the specimen it is going to be used on. Reach for the folder only for a structure meant as a better generic default than the published one — a composition typical of the material a laboratory usually sees — and name it so it says which.

## What it reads, and what it reports

A structure library kept in a laboratory is written partly by hand and partly by TOPAS, and the two differ. The import reads both: a value carrying the backtick TOPAS puts on a refined parameter (`16.875164``, or with a limit welded on as `9.663417`_LIMIT_MIN_9.5`); an occupancy given as an expression with the value TOPAS last evaluated after a colon (`occ Fe+2 =FeMgOCC;:0.99996``), which keeps a split site split instead of putting a full atom on each half; a space group written with spaces (`"C c c m"`); a comment after `str`; and a second structure block that takes its cell from the first by name (`a =a_Calcite;`), which is how a two-component phase for a crystallinity distribution is written.

**Anything not imported is named.** A phase that cannot be read is listed with the reason — no cell, no space group, no sites — and a phase that lists reflections rather than atoms (`hkl_Is`, a Pawley or amorphous entry) is named as such, since ClayQuant calculates from structures and has nothing to calculate from. Items that are headings or helper macros are passed over silently, as a menu is full of them. Before this, an item that could not be read was dropped before it became a record, so it was absent from every count afterwards: adding a phase to the library and re-importing gave the same number of phases as before, with nothing to say why.

## 4.4 Building the reference library

A library is built once for a given instrument configuration and measurement grid, and reused. Build it on the same 2θ grid as the measurements.

```
clayquant-build-library -o library/clays.npz
```

Building the default library takes about three minutes. Within the interface the same operation is available as *Build library now*, using the current instrument settings.

## Chlorite compositions

The library holds one chlorite by default, the published structure, and that is the recommended setting. The two octahedral sheets of a chlorite take iron for magnesium in proportions that vary from one deposit to the next, iron scatters about twice as strongly as the magnesium it replaces, and each sheet sits at a definite height in the layer, so the composition does act directly on the basal intensities a quantification is read from. Whether that matters enough to fit is a separate question, and on the two pure chlorites of §A.17 it did not: the fit improved the residual only 1.5 to 1.7 times, drove one specimen to implausibly little iron, and ran against the bound of the crystallite-thickness grid.

So the facility is there for a specimen where the evidence calls for it, and is off by default. Unlike the illite/smectite ratio there is no generic set of compositions worth spanning; fit the composition to a measured pattern of a pure chlorite, check that the residual improves by enough to be worth having and that nothing lands on a bound, and only then calculate the library for the values that came out:

```
clayquant-build-library --chlorite-iron 0.35/0.20,0.55/0.30
```

Each pair is the iron occupancy of the 2:1 layer's octahedral sheet and of the interlayer hydroxide sheet. Every pair gets its own set of chlorite patterns, all grouped under the phase `chlorite`, so the fit chooses among compositions as it already chooses among layer spacings and orientations, and the quantification reports one chlorite. The cell mass follows the substitution, so the weight percent is consistent with whichever composition the fit chooses.

To measure the pairs, from a pattern of the pure mineral:

```
from clayquant.composition import fit_chlorite_iron
from clayquant.io import read_pattern

fit = fit_chlorite_iron(read_pattern("Chlorite_pure.xrdbl"), instrument)
print(fit.describe())
```

The fit uses ratios of basal intensities within the one pattern, not the intensities themselves, so it needs neither a standard nor a known amount of material nor a known texture: preferred orientation multiplies a basal series by a single factor and divides straight out. It reports the residual beside the residual of the published structure, so a composition that is no improvement says so — and reports the crystallite thickness it settled on, because one at the edge of the grid means the composition absorbed something that was not composition.

Check the integration window before believing the result. A clay basal reflection has a diffuse tail from stacking disorder that carries much of its area, so a window that looks ample beside the peak width can hold a different fraction of each order — which is exactly what a ratio of orders responds to. §A.17 records a conclusion drawn from a window too narrow by a factor of three, and its retraction.

# 5 Operating procedure

## 5.1 Specimen preparation assumed

ClayQuant assumes the preparation customary for clay mineral analysis: the bulk sample is ground, the clay fraction separated by sedimentation or centrifugation, and the separate deposited on glass slides to give oriented mounts, of which one is left air-dried, one solvated with ethylene glycol and one heated, typically to 500 °C. A bulk (randomly oriented) measurement of the whole sample is useful context but is not required by the program.

## 5.2 Step 1 — Load

Enter the folder holding the measurements and press *Scan folder*. The readable files are listed in the three drop-down menus. Choose *one* of the three mounts: the other two are then filled in automatically, because the three scans of a sample normally differ in a single token of the file name, so `DBB_17_PW_air.xrdbl` identifies `DBB_17_PW_eg.xrdbl` and `DBB_17_PW_heat.xrdbl`. It works from whichever mount is chosen first and in whichever format the files are held; a line beneath the menus reports what was matched, and says so when a mount is missing from the folder. Press *Load*.

The tokens recognised are `air`, `airdried`, `airdry`, `ad` and `lutro` for the air-dried mount; `eg`, `egl`, `gly`, `glyc`, `glycol` and `ethylenglykol` for the glycol-solvated one; and `heat`, `heated`, `ht`, `350`, `500` and `550` for the heated one. A token counts only where it stands as a word of its own, delimited by an underscore, hyphen, dot or space or ending the name — otherwise the `ad` of `Bad_Segeberg` would name an air-dried mount — and the rest of the name must agree exactly, so `DBB_17_air` finds `DBB_17_eg` and never `DBB_17_PW_eg`. Files of the format chosen are preferred, so a folder holding both XRDML and `.xy` exports of everything does not yield one of each. Any assignment can be overridden by hand, and a box already holding a mount of the same sample is left untouched.

**Paths under WSL.** If ClayQuant runs inside WSL while the measurements sit on the Windows side, a path copied from Explorer such as `C:\Users\you\samples\TM` refers to a drive Linux does not have: Windows drives appear under `/mnt`, so that folder is `/mnt/c/Users/you/samples/TM`. Either form may be typed — a drive letter is translated automatically, surrounding quotes are stripped, and a `\\wsl$\Distro\home\you` share is mapped back to `/home/you`. The folder that was actually opened is named in the confirmation, so it is visible which one was used.

The point count, angular range and step are reported for each mount. Check that all three cover the same range: the diagnostics compare mounts point by point after interpolation, and a mount measured over a different range will still work but will restrict the comparison.

## 5.3 Step 2 — Zero error

The  $2\theta$  zero error is determined on the quartz 100 reflection,  $d = 4.2551 \text{ \AA}$ ,  $2\theta = 20.859^\circ$  for Cu  $K\alpha_1$ . This reflection is chosen because no clay basal reflection falls on it.

*Find peak automatically* locates it and sets the slider; the slider itself steps in  $0.01^\circ$ , half the usual measurement step, and the merit curve beneath shows the figure of merit against candidate offsets so the chosen value can be seen to be the optimum. Each mount carries its own zero error, because each is a separately prepared slide with its own displacement; set all three.

**Quartz 100 is only 0.4° from the kaolinite 020 band at 4.36 Å.** In a clay separate quartz is often weak while kaolinite is strong, and a wide search window will find kaolinite and return a zero error of about  $-0.4^\circ$ , displacing the whole pattern. The search window is therefore deliberately narrow ( $\pm 0.30^\circ$  by default) and a signal-to-noise test is applied. Where quartz is genuinely absent the program reports that it found no calibration peak and leaves the shift at zero rather than returning a meaningless number; set the value by hand or calibrate on another reflection. See §A.5.

## 5.4 Step 3 — Background

Choose a model and adjust it until it follows the background without cutting into the peaks. The components are described in §6; the  $1/x$  term, which describes the low-angle rise from air scatter and the direct beam, can be added to any of the others. A non-parametric peak-stripped background is also available for backgrounds no low-order function follows.

Models are fitted not to the raw counts but to a peak-stripped estimate of the background, then refined asymmetrically. This matters: a clay pattern has no peak-free points below about  $8^\circ$ , so fitting the raw intensities there drags the background up into the basal reflections and removes the very signal being quantified. Peak stripping errs the other way, underestimating the background beneath a broad strong low-angle reflection, which is the safe direction but means the low-angle background should always be checked by eye.

*Apply to all mounts* uses the same model for the three mounts, which is appropriate when they were measured under identical conditions.

## 5.5 Step 4 — Main minerals

Load the phase database of §4.3 and press *Search for main minerals*. If a reference library has been loaded, every candidate is calculated and fitted in competition with the clay patterns and with each other in a single non-negative least-squares solve, and each is ranked by the share of the pattern it takes. Without a library, the program falls back to matching predicted peak positions, which is a much weaker test (§A.6).

A dialog lists what was found:

*Evidence for the following main minerals has been found in the data:*

with, for each, its share of the pattern, how many of its expected lines were found, and the best signal-to-noise among them. Phases with a clear share are ticked; the tail at one or two per cent is listed but left unticked, being largely least-squares slack rather than mineralogy.

**This is a shortlist for judgement, not an identification.** Phases with similar cells — the feldspars, the spinels, dolomite against ankerite — cannot be separated on diffraction evidence alone at these levels, and a database assembled for general laboratory use contains many phases that cannot occur in a sediment. Tick what the sample can plausibly contain. Selected phases are fitted with the clays and reported separately, then removed when the clay percentages are normalised.

## 5.6 Step 5 — Kaolinite

Kaolinite is identified by the loss of its 7.15 Å reflection on heating. Because the two mounts are separate slides with different amounts of material, the comparison is meaningless until they are put on a common scale; the program does this on a reflection that survives heating, by default the illite 10 Å reflection, and reports the scale factor used.

The result is the percentage of the 7.15 Å area lost. What survives is chlorite 002, which occupies the same position and does not collapse. A sample with both minerals therefore shows partial collapse, and the surviving fraction is an estimate of the chlorite contribution to that peak.

## 5.7 Step 6 — Expandable clay

Expandable layers are identified by the shift of the low-angle reflection on glycol solvation: a smectite interlayer takes up a two-layer glycol complex and expands to about 16.9 Å. The program locates the reflection in a window of the air-dried and glycolated mounts, reports both positions in ångström and the shift between them, and the ratio of the areas.

A shift towards larger  $d$  is the diagnostic. A shift of 13.8 → 16.1 Å, for example, is unambiguous evidence of expandable layers. No shift means either that there are none, or that the expandable component is a minor interstratified fraction whose effect is spread through the illite/smectite series rather than concentrated in a discrete peak — which the quantitative fit will still detect.

## 5.8 Step 7 — Fit and results

Load a reference library, choose the mount to fit — normally the glycol-solvated one — set the angular range, and press *Run NNLS fit*.

Choose the fitting range deliberately. The lower limit should exclude the direct-beam tail, and 4° is a reasonable default; setting it lower rarely adds information and can let the fit chase background. The upper limit should include the region where the main minerals have their strong lines, so that they are properly determined, but there is little clay information above about 35°.

*Restrict orientation parameters* limits the fit to library patterns of particular March–Dollase values. Since a well-prepared oriented mount is strongly textured, restricting to the low values (0.1–0.3) both speeds the fit and prevents it from using near-random patterns to model what is really a texture effect.

The results appear as four plots and two tables, described in §8. *Export tables and plots* writes them to disk.

**Read the plots, not only the table.** The fit will always return numbers. Whether they mean anything is visible in the difference curve of the *Fit* plot and in whether the per-phase curves of the *All phases* plot sit under real features of the measurement. A large systematic residual at one strong peak usually means a layer spacing or crystallite thickness outside the range the library spans (§9).

## 5.9 The simulator

The simulator drives the calculation directly, independently of any measurement, for teaching, for exploring what a given composition looks like, and for checking the effect of a parameter. It computes a single interstratified or discrete pattern from the controls of §6 and displays it immediately.

# 6 Reference: interface controls

Every control is listed below by the tab on which it appears, with the identifier used internally, its default, and its meaning. Sliders show their current value; range sliders have two handles.

## 6.1 Tab 1 — Load

| Control                              | Identifier                         | Default | Meaning                                                                                                                                                                                                                |
|--------------------------------------|------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data folder                          | directory                          | —       | Folder searched for measurement files. Sub-folders are not searched.                                                                                                                                                   |
| Scan folder                          | scan                               | —       | Lists the readable files and offers them in the three drop-downs. Sub-folders are not searched.                                                                                                                        |
| Air-dried / Ethylene glycol / Heated | file-air, file-glycol, file-heated | —       | Assignment of a file to each mount. Choosing one fills the other two from the file names (§5.2); either can then be changed by hand. The glycol mount is the one normally fitted; the other two drive the diagnostics. |
| Matched files                        | file-pairing                       | —       | Reports which files were assigned automatically, and names any mount for which no matching file is present in the folder.                                                                                              |
| Load                                 | load                               | —       | Reads the selected files and reports points, range and step for each.                                                                                                                                                  |

## 6.2 Tab 2 — Zero error

| Control                         | Identifier     | Default              | Meaning                                                                                                                                                        |
|---------------------------------|----------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mount                           | zero-mount     | Air-dried            | Which mount the slider and the automatic search act on. Each mount has its own zero error.                                                                     |
| Calibration reflection          | zero-reference | Quartz 100, 4.2551 Å | The reflection whose known position defines the zero. Quartz 100 clears all clay basal reflections. Quartz 101 (3.3435 Å) is stronger but overlaps illite 003. |
| Zero error (°2θ, steps of 0.01) | zero-slider    | 0.00                 | The offset <i>subtracted</i> from the measured angles. The step is 0.01°, half the usual 0.02° measurement step.                                               |
| Find peak automatically         | zero-auto      | —                    | Locates the calibration peak within a narrow window and sets the slider. Reports the signal-to-noise, or states that no calibration peak was found.            |

## 6.3 Tab 3 — Background

| Control                                   | Identifier             | Default   | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------|------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mount                                     | bg-mount               | Air-dried | Mount displayed and fitted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Main component                            | bg-kind                | Chebyshev | <i>Polynomial</i> : $\Sigma c_k u^k$ in the angle scaled to $[-1,1]$ . <i>Chebyshev</i> : $\Sigma c_k T_k(u)$ , better conditioned at high degree and the usual Rietveld choice. <i>Exponential</i> : a single decaying term. <i>Peak-stripped</i> : non-parametric; the stripped estimate is used directly and follows any shape, at the cost of having no parameters.                                                                                                                                                                                                                        |
| Degree                                    | bg-degree              | 4         | Degree of the polynomial or Chebyshev component; degree $n$ contributes $n+1$ terms. Raise it only as far as the background demands: an over-flexible background follows peaks.                                                                                                                                                                                                                                                                                                                                                                                                                |
| Exponential decay (1/°)                   | bg-decay               | 0.20      | Decay constant of the exponential component, in reciprocal degrees.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Add 1/x term                              | bg-inverse             | on        | Accumulates $c/(2\theta+x_0)$ on top of the main component. This describes the low-angle rise from air scatter and the direct beam and can be combined with any of the others.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 1/x offset (°2 $\theta$ )                 | bg-offset              | 1.0       | The offset $x_0$ , which keeps the term finite as the angle approaches zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Peak-stripping width (°2 $\theta$ )       | bg-snip                | 4.0       | Width of the widest feature the stripping treats as a peak. The components are fitted to the stripped estimate and follow it to a few counts, so this control, not the choice of component, is what decides where the background lies. Set it just above the width of the broadest reflection to be kept: narrower leaves peak wings standing in the background and eats into the reflections. It does <i>not</i> control the low-angle tail, which is convex and is therefore preserved at any width (§A.11). Measured over eleven mounts the fit flattens by about 4°, which is the default. |
| Apply to this mount / Apply to all mounts | bg-apply, bg-apply-all | —         | Stores the fitted background. <i>All</i> uses the same model for the three mounts, appropriate when they were measured under identical conditions.                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Reading the plot.** Four curves are drawn: the *measured* pattern in grey, the *background model* as a thick orange line, the *peak-stripped estimate* as a blue dashed line drawn over it, and the *subtracted* result in green. The dashes should lie on the orange line; where they part, the chosen components cannot follow the shape of the background and the degree or the component should be changed. The blue curve is the quantity the stripping-width slider controls, and the one the model is fitted to.

**Reading the status line.** It reports how closely the model follows the stripped estimate, in counts, overall and over the lowest degree separately, and then what the model leaves at the low-angle end: the background value at the first measured point, the measurement there, and the difference as a percentage. That difference is the intensity the fit will be asked to explain as diffraction, and on a scan that opens on the direct-beam tail it should be small - a few percent. A large share there means the background is not following the tail. A goodness-of-fit figure is deliberately *not* reported for the background: it is close to unity for almost any choice, because the estimate is smooth and the model has several free terms, and it would read as success while the background sat a thousand counts below the measurement at 3°.

6.4 Tab 4 — Main minerals

| Control                               | Identifier                    | Default                | Meaning                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------|-------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phase database (.json)                | db-path                       | structures/phases.json | The database produced by clayquant-import-structures.                                                                                                                                                                                                                                                                                              |
| Load database                         | db-load                       | —                      | Reads it and reports how many phases it holds. Phyllosilicates in the database are left to the clay library and are not offered here.                                                                                                                                                                                                              |
| Mount to search                       | detect-mount                  | Air-dried              | Which mount is searched. The air-dried mount is usually the best choice: glycol dilutes the accompanying minerals and heating alters some of them.                                                                                                                                                                                                 |
| Unit cell allowance (%)               | detect-allowance              | 2                      | Relative cell deviation allowed when matching positions, applied as one <i>coherent</i> scaling of all of a phase's d-spacings. Most accompanying minerals are solid solutions whose cells differ from the database entry. Applied per peak instead of coherently, a 2 % allowance is a window up to 1.6° wide and matches almost anything (§A.6). |
| Minimum share (‰) or score (%)        | detect-score                  | 10                     | Reporting threshold. With a library loaded it is read as parts per thousand of the pattern (10 = 1 ‰); without one, as a percentage score of the peak-matching test.                                                                                                                                                                               |
| Minimum peak S/N                      | detect-snr                    | 5                      | Height above the local noise a predicted line must reach to count as found. Used by the peak-matching fallback and by the per-line evidence shown in the dialog.                                                                                                                                                                                   |
| Search range (°2θ)                    | detect-range                  | 4 to 40                | Angular range considered. It should cover the strong lines of the expected accompanying minerals.                                                                                                                                                                                                                                                  |
| Search for main minerals              | detect-run                    | —                      | Runs the search and opens the confirmation dialog.                                                                                                                                                                                                                                                                                                 |
| Include selected in the fit / Dismiss | detect-accept, detect-dismiss | —                      | Confirms the tick list, or discards it. Nothing is added to the fit until confirmed.                                                                                                                                                                                                                                                               |

6.5 Tab 5 — Kaolinite

| Control                | Identifier    | Default      | Meaning                                                                                                                                             |
|------------------------|---------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.15 Å window (°2θ)    | kao-window    | 11.6 to 13.2 | Integration window for the kaolinite 001 / chlorite 002 reflection.                                                                                 |
| Scaling reference      | kao-reference | Illite 10 Å  | The reflection used to put the air-dried and heated mounts on a common scale. It must survive heating. <i>Manual</i> uses the factor below instead. |
| Manual scale factor    | kao-scale     | 1.00         | Used when the scaling reference is set to manual.                                                                                                   |
| Compare air and heated | kao-run       | —            | Reports the percentage of the 7.15 Å area lost, the surviving fraction attributable to chlorite 002, both areas, and the scale factor used.         |

## 6.6 Tab 6 — Expandable

| Control                | Identifier    | Default     | Meaning                                                                                                                                                                              |
|------------------------|---------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Search window (°2θ)    | sme-window    | 4.0 to 10.0 | Window searched for the low-angle reflection in both mounts. It must be wide enough to hold the air-dried position and the glycolated position, which may differ by several degrees. |
| Scaling reference      | sme-reference | Illite 10 Å | As for kaolinite: puts the two mounts on a common scale so that the area ratio is meaningful.                                                                                        |
| Manual scale factor    | sme-scale     | 1.00        | Used when scaling is manual.                                                                                                                                                         |
| Compare air and glycol | sme-run       | —           | Reports both positions in ångström, the shift, and the area ratio.                                                                                                                   |

## 6.7 Tab 7 — Fit and results

| Control                                 | Identifier              | Default           | Meaning                                                                                                                                                                                                              |
|-----------------------------------------|-------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Library file (.npz)                     | lib-path                | library/clays.npz | The stored reference library.                                                                                                                                                                                        |
| Load library / Build library now        | lib-load, lib-build     | —                 | Loads a stored library, or builds one with the current instrument settings. Building takes a few minutes.                                                                                                            |
| Mount to fit                            | fit-mount               | Ethylene glycol   | Which mount is quantified. The glycolated mount is the standard choice because glycolation resolves the illite/smectite series.                                                                                      |
| Fit range (°2θ)                         | fit-range               | 4 to 34           | Range fitted. Exclude the direct-beam tail at the low end; there is little clay information above about 35°.                                                                                                         |
| Restrict orientation parameters         | fit-orientations        | all               | Limits the fit to library patterns of the chosen March–Dollase values. Restricting to low values is appropriate for a well-oriented mount and prevents the fit using near-random patterns to model a texture effect. |
| Subtract the fitted background          | fit-subtract            | on                | Subtracts the background stored in step 3. Turn it off only if the measurement is already background-corrected.                                                                                                      |
| Run NNLS fit                            | fit-run                 | —                 | Performs the weighted non-negative least-squares fit and produces the plots and tables.                                                                                                                              |
| Export folder / Export tables and plots | export-path, export-run | results           | Writes the outputs of §8.                                                                                                                                                                                            |

## 6.8 Simulator

| Control                                    | Identifier    | Default         | Meaning                                                                                                                   |
|--------------------------------------------|---------------|-----------------|---------------------------------------------------------------------------------------------------------------------------|
| Model                                      | sim-model     | Interstratified | <i>Interstratified</i> computes a two-component stack; <i>Discrete phase</i> computes the pure phase from its structure.  |
| Phase / host                               | sim-phase     | illite          | The discrete phase, or the host layer type A of the interstratification (the other component being glycol–smectite).      |
| Host layer fraction                        | sim-fraction  | 0.80            | $W_A$ , the proportion of host layers. 1.0 is the pure host; 0.0 the pure smectite.                                       |
| Junction probability<br>P(host → smectite) | sim-pab       | 0 (random)      | $P_{AB}$ of §2.7. Zero selects random stacking ( $R = 0$ ); values above $W_B$ give ordering, below it segregation.       |
| Mean crystallite thickness<br>(layers)     | sim-csds      | 10              | $\langle N \rangle$ of the lognormal CSDS. This is the principal control on basal peak width.                             |
| CSDS width $\beta$ (in $\ln N$ )           | sim-beta      | 0.35            | Width of the distribution in $\ln N$ . Larger values give a broader spread of thicknesses and smoother peak profiles.     |
| March–Dollase orientation<br>$r$           | sim-po        | 0.3             | Texture parameter. Recall that it does not change the shape of a purely basal series, only the basal-to-non-basal ratio.  |
| Glycolated smectite d(001)<br>(Å)          | sim-dsmectite | 16.86           | Repeat distance of the glycol–smectite layer. 16.86 Å is the value measured by Reynolds (1965); 16.9–17.1 Å is also used. |
| 2 $\theta$ range (degrees)                 | sim-range     | 2 to 40         | Calculated range.                                                                                                         |
| Options                                    | sim-options   | —               | Toggles for including the non-basal contribution and for showing the component curves separately.                         |

## 7 Reference: command line

There are three commands, and three ways to reach each of them. The `./clayquant` script in the project directory (`clayquant.ps1` on Windows) is the one that needs nothing set up: it locates the virtual environment itself and runs the module, so it works whether or not the environment is activated and whether or not the installed command was ever written.

| From the project directory                 | With the environment activated           | As a module                              |
|--------------------------------------------|------------------------------------------|------------------------------------------|
| <code>./clayquant build-library</code>     | <code>clayquant-build-library</code>     | <code>python -m clayquant.library</code> |
| <code>./clayquant import-structures</code> | <code>clayquant-import-structures</code> | <code>python -m clayquant.bern</code>    |
| <code>./clayquant gui</code>               | <code>clayquant-gui</code>               | <code>python -m clayquant.gui.app</code> |

Options are identical in all three forms; the tables below list them once.

**“No module named clayquant”, or a command not found.** Both mean the shell is not using the virtual environment. A bare python is then the system interpreter, which has never heard of ClayQuant, and the installed commands are not on PATH. Either activate the environment (source `.venv/bin/activate`) or use `./clayquant`, which does not need it. The error names the interpreter it used — `/usr/bin/python` rather than something under `.venv` is the symptom.

**A second reason a command can be missing.** The installed commands are written into the environment when the package is installed, and at no other time. Updating the source — `git pull` — therefore does not create a command that the new version has added: the environment keeps the set it was built with, and the new command is reported as not found although its code is present in the checkout. Running `./install.sh` (or `install.ps1`) again writes them all; it lists every command it has checked and names any that is missing rather than reporting success. `./clayquant` is unaffected either way, since it runs the module rather than the command.

## 7.1 clayquant-build-library

| Option                                        | Default                        | Meaning                                                                 |
|-----------------------------------------------|--------------------------------|-------------------------------------------------------------------------|
| <code>-o, --out</code>                        | <code>library/clays.npz</code> | Output file.                                                            |
| <code>--two-theta-min, --two-theta-max</code> | <code>2, 40</code>             | Angular range of the calculated grid, in degrees.                       |
| <code>--step</code>                           | <code>0.02</code>              | Step of the calculated grid. Match the measurement.                     |
| <code>--csds-means</code>                     | <code>5 15 50</code>           | Mean crystallite thicknesses, in layers; one set of patterns per value. |
| <code>--csds-beta</code>                      | <code>0.35</code>              | Width of each lognormal distribution in $\ln N$ .                       |
| <code>--orientations</code>                   | <code>0.1 ... 1.0</code>       | March–Dollase parameters to span.                                       |
| <code>--quiet</code>                          | <code>off</code>               | Suppress progress reporting.                                            |

## 7.2 clayquant-import-structures

| Option                 | Default                             | Meaning                                               |
|------------------------|-------------------------------------|-------------------------------------------------------|
| <code>xml</code>       | <code>—</code>                      | The jEdit macro-menu XML file to import (positional). |
| <code>-o, --out</code> | <code>structures/phases.json</code> | Output phase database.                                |
| <code>--quiet</code>   | <code>off</code>                    | Suppress the per-phase notes and the failure list.    |

## 7.3 clayquant-gui

| Option       | Default   | Meaning                                                                                                                                                                                                            |
|--------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --port       | 8050      | Port to serve on. If it is already in use - a second copy started from the icon, most likely - the next free port is taken instead and reported, rather than the program failing where no one can see the message. |
| --host       | 127.0.0.1 | Interface to bind. The default is the loopback, so the interface is not reachable from the network.                                                                                                                |
| --debug      | off       | Enable the development server's reloader and error pages.                                                                                                                                                          |
| --browser    | on        | Open the interface in the default browser once the server is answering. This is what makes the desktop icon work: it is started from a double-click, with no terminal to read an address from.                     |
| --no-browser | —         | Serve without opening anything, for a script or a machine being used remotely.                                                                                                                                     |

## 8 Output files

*Export tables and plots* writes seven files into the chosen folder.

| File                       | Contents                                                                                                                                            |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| clayquant_mounts.html      | The three mounts overlaid: air-dried, glycol-solvated, heated. The plot in which the kaolinite collapse and the glycol expansion are seen directly. |
| clayquant_fit.html         | The measurement, the fitted curve and the difference, with $R_{wp}$ in the title.                                                                   |
| clayquant_phases.html      | The measurement with every fitted phase drawn separately, and their sum.                                                                            |
| clayquant_clay_phases.html | The same for the clay minerals alone, annotated with the clay share of the whole pattern.                                                           |
| clayquant_all_phases.csv   | Quantification over all fitted phases.                                                                                                              |
| clayquant_clays_only.csv   | Quantification over the clay minerals alone, renormalised to 100 %.                                                                                 |
| clayquant_curves.csv       | The numerical curves: $2\theta$ , measured, fit, and one column per phase, so the figures can be redrawn in any other program.                      |

The figures are self-contained HTML: they need no plotting software to open and keep the zoom and hover of the interactive versions.

## 8.1 Columns of the quantification tables

| Column             | Meaning                                                                                                                                                                                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phase              | Phase name. Interstratified series appear as I/S and C/S, summed over the individual library patterns that contributed.                                                                                                                                                                                                                 |
| group              | clay or main. Determines which phases the clay-basis normalisation keeps.                                                                                                                                                                                                                                                               |
| coefficient        | Sum of the fitted scale factors $c_p$ of that phase's patterns. Depends on the arbitrary normalisation of the library and is meaningful only in comparison within one fit.                                                                                                                                                              |
| scattering_percent | The phase's share of the calculated integrated intensity. <b>The primary number.</b>                                                                                                                                                                                                                                                    |
| amplitude_percent  | The phase's share of the summed scale factors.                                                                                                                                                                                                                                                                                          |
| entries            | The individual library patterns that took non-zero coefficients, naming the composition, orientation parameter, crystallite thickness and layer spacing selected. This is where the fit reports what it found: an I/S share carried by 0.90/0.10 patterns at $d = 9.95 \text{ \AA}$ and $N = 50$ is a specific mineralogical statement. |

Each file carries a header recording the measurement,  $R_{wp}$  and  $R_p$ , the basis of normalisation, the clay share of the whole pattern where applicable, and a statement that the percentages are shares of diffracted intensity and not weight percent, so that a file cannot later be mistaken for a mass analysis.

## 9 Limitations

These are properties of the method, not defects to be worked around. Each is stated with its practical consequence.

1. **A fixed library cannot refine.** Peak positions, widths and orientation are sampled at the values the library was built with. A specimen whose layer spacing or crystallite thickness falls outside the spanned range will fit badly, and the misfit appears as a systematic residual at a strong peak rather than as an obviously wrong answer. Widen the spanned ranges and rebuild if this is seen.
2. **Percentages are shares of intensity, not of mass** (§2.13).
3. **Kaolinite polytypes are not distinguishable from basal reflections** (§2.3). A division of intensity between  $1M$  and  $2M$  is not a polytype determination.
4. **The non-basal contribution of an interstratified phase is approximated** by that of its host (§2.12). Turbostratic  $hk$  bands are not modelled with their true sawtooth profile. The approximation is good for a strongly oriented mount and degrades as  $r \rightarrow 1$ .
5. **Only two-component interstratifications are treated.** Three-component systems, and mixed-layer chlorite/smectite/illite, are outside the model.
6. **Accompanying minerals of similar cell cannot be separated.** The screen ranks candidates by explained intensity, which is far sharper than position matching, but the feldspars, the spinels and the rhombohedral carbonates remain close enough that the ranking should be read as a shortlist for judgement (§5.5).
7. **The zero error requires quartz.** Where quartz is absent the program declines to report a value; it must then be set by hand, and an error there displaces the whole pattern and corrupts the fit.
8. **The low-angle background is the weakest part of the reduction.** There are no peak-free points below about  $8^\circ$ , so the background there is an extrapolation. It should always be inspected by eye, particularly when a smectite 001 near  $5.2^\circ$  is at

issue.

9. **The ordered case is implemented but not spanned by the default library** (§2.7). Regularly interstratified minerals such as rectorite and corrensite will not be fitted correctly by a library built with the default settings.
10. **No absorption or microabsorption correction is applied.** For an infinitely thick specimen in Bragg–Brentano geometry the absorption factor is independent of angle and cancels in relative intensities; a thin mount violates this, as does a specimen of strongly contrasting absorption between phases.

## 10 References

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The list below covers the physical model, the numerical methods, the software libraries, and the four crystal structures on which ClayQuant's clay mineral calculations rest. The accompanying mineral structures come from a user-supplied library and are not covered here, because their citations belong to the database they were drawn from; every one of them is identified by database and entry number in Appendix D, from which the citation can be retrieved.

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## Appendix A — Record of verification tests

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This appendix records the tests by which ClayQuant was verified during development: what each test was for, how it was done, what it returned, and what was concluded. Several tests are recorded that *failed* and led to a change in the program; these are the

more informative ones and are reported in full.

**On the measurements used.** Development was verified against 33 measurements of eleven clay separates from a working laboratory. Those measurements are proprietary and are **not** reproduced here, nor distributed with the program. They are referred to below as Specimen 1, Specimen 2 and so on, and only derived quantities — agreement factors, peak widths, fitted positions — are quoted, never the measured data themselves.

## A.1 The interstratification model against closed-form results

**Reasoning.** The matrix method of §2.5 is the core of the program and is easy to implement subtly wrongly: the phase convention of equation (2.10) can be transposed, the pair counting of equation (2.11) can be off by one, and the stationary distribution can be inconsistent with the transition matrix. None of these produce obviously absurd output — they produce a plausible pattern that is wrong. The model must therefore be tested against cases whose answer is known independently.

**Method and results.** Twenty-five tests (`test_mixed_layer.py`):

| Test                                                                                                               | Reasoning                                                                                                                                                                                                                                                                       | Result                                                                 |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Single component reproduces the Laue interference function, for $N = 1, 2, 5, 17$                                  | With one layer type the general result must collapse to the closed form (2.7). This checks the phase convention, the pair counting and the $N$ -dependence simultaneously, against an exact expression.                                                                         | Agreement to numerical precision at all four thicknesses. <b>Pass.</b> |
| Two identical layer types behave as one                                                                            | Setting A and B to the same layer must give the one-component answer whatever the proportions, which tests that the two components are combined without double counting.                                                                                                        | Identical to the single-component calculation. <b>Pass.</b>            |
| A pure end member ignores the absent layer                                                                         | At $W_A = 1$ the pattern must not depend on what B is. A failure here would show the absent component leaking through the transition matrix.                                                                                                                                    | Independent of the B layer model. <b>Pass.</b>                         |
| Single-layer crystallites show no interference                                                                     | At $N = 1$ the intensity must be the layer structure factor squared, with no interference term at all. This isolates the pair sum.                                                                                                                                              | Reduces to $\sum W_m  F_m ^2$ . <b>Pass.</b>                           |
| Intensity is non-negative — 15 cases: random, ordered and segregated stacking at $W_A = 0.05, 0.2, 0.5, 0.8, 0.95$ | Intensity is a squared modulus and cannot be negative anywhere. A sign error in the real part of the matrix sum, or an inconsistent stationary distribution, shows up as small negative excursions between peaks. This is the most sensitive cheap test of the whole formalism. | Non-negative throughout in all 15 cases. <b>Pass.</b>                  |
| CSDS pair weights                                                                                                  | $G(n)$ of equation (2.14) must equal 1 at $n = 0$ and decrease monotonically to zero, or the thickness distribution is misapplied.                                                                                                                                              | As specified. <b>Pass.</b>                                             |
| The random transition matrix is the Markov special case                                                            | Building a matrix with $P_{AB} = W_B$ through the general Markov constructor must give exactly the random one. This checks the constructor against its own special case.                                                                                                        | Identical matrices. <b>Pass.</b>                                       |
| Impossible ordering is rejected                                                                                    | A $P_{AB}$ implying a probability outside $[0,1]$ is unphysical and must raise rather than silently produce nonsense.                                                                                                                                                           | Raises. <b>Pass.</b>                                                   |

**Conclusion.** The interstratification model reproduces every case for which an independent answer exists, and satisfies the positivity constraint across the full range of composition and ordering.

## Consistency of the basal and three-dimensional calculations

**Reasoning.** The same phase can be computed two ways — as a three-dimensional structure through equation (2.3), or as a one-dimensional layer stack through §2.5. These are independent code paths that must agree on the basal series of a pure phase, otherwise the mixed-layer patterns and the discrete patterns are on different scales and the fit apportions intensity wrongly between them.

**Results** (`test_basal_consistency.py`): for illite, chlorite, kaolinite 1*M* and kaolinite 2*M*, the basal series computed from the one-dimensional layer model matches the 00*l* intensities of the full three-dimensional calculation; the derived layer thickness equals *d*(001) of the cell divided by the number of layers per cell; and the absolute spacings agree with the accepted values (10.02, 14.28, 7.15, 7.15 Å). **Pass.**

**Conclusion.** The polytype folding of §2.3 is exact for the structures used, and the two code paths are on a common scale.

## A.2 Validation against Reynolds (1965)

**Reasoning.** Everything above is internal consistency: the program agreeing with itself. An external check against a published experiment is needed, and the glycol–smectite complex is the ideal one, because Reynolds (1965) published both his structural model (his Table 1) and his observed and calculated structure factors and intensities for fourteen basal orders (his Table 2). Reproducing his calculated quantities from his stated model tests the layer structure factor of equation (2.6), the scattering factors, the displacement factors and the Lorentz–polarization treatment together, against data that did not come from this program.

**Method.** The model of his Table 1 was transcribed, the basal structure factors computed from equation (2.6), converted to intensities with the random-powder Lorentz–polarization factor he states he used, normalised at 001, and compared with his Table 2. Agreement is measured by his own reliability factor,  $R = \Sigma ||F_o| - |F_c|| / \Sigma |F_o|$ .

**An ambiguity, and how it was resolved.** His Table 1 gives occupancies per atomic plane, but the octahedral plane lies on the mirror at *z* = 0 and is quoted as half its content; and his glycol rows are labelled as half-molecules ("H<sub>2</sub>C–HO"), which may or may not include their hydrogens. Four readings are possible. Rather than guess, all four were computed and compared with his published values:

| Reading                                            | Electrons per cell | <i>R</i> (  <i>F</i> <sub>o</sub>  ) | Mean  Δlog <sub>10</sub> |
|----------------------------------------------------|--------------------|--------------------------------------|--------------------------|
| A — octahedral plane as quoted, glycol without H   | 452                | 0.299                                | 0.350                    |
| B — octahedral plane as quoted, glycol with H      | 476                | 0.312                                | 0.327                    |
| C — octahedral plane doubled, glycol without H     | 480                | 0.077                                | 0.158                    |
| <b>D — octahedral plane doubled, glycol with H</b> | <b>504</b>         | <b>0.039</b>                         | <b>0.028</b>             |

**Result.** Reading D is correct by a wide margin. It also reproduces independently the electron count Reynolds states for his cell ("about 500 electrons"): 504 against his ~500, where the other readings give 452 to 480. Its reliability factor, *R* = 0.039, is slightly *better* than the *R* = 0.042 he reports for his own best model — as expected, since modern scattering factors are used. Four tests now assert this permanently (`test_reynolds_1965.py`): the electron count, the reliability factor, the fourteen calculated intensities, and the ordering of the strong orders.

**Conclusion.** The one-dimensional structure factor calculation reproduces a published experimental analysis to better than its original agreement. This is the strongest external validation in the program, and it simultaneously confirms the choice of Lorentz–

polarization factor of §2.8, since no other choice reproduces these intensities.

### A.3 Import of a TOPAS structure library

**Reasoning.** The accompanying minerals enter the fit as calculated patterns, so an error in reading their structures — a misparsed cell parameter, a wrong space group setting, a mis-assigned occupancy — produces a wrong pattern that the fit will nevertheless use. The import must be verified against known diffraction data, phase by phase.

**Method and results.** A library of 205 phases was imported and the strongest calculated line of each of sixteen well-known phases compared with its accepted position.

| Phase      | Calculated (°2θ) | Accepted (°2θ) | Δ     |
|------------|------------------|----------------|-------|
| Quartz     | 26.64            | 26.64          | +0.00 |
| Calcite    | 29.40            | 29.40          | +0.00 |
| Rutile     | 27.44            | 27.44          | +0.00 |
| Magnetite  | 35.43            | 35.44          | −0.01 |
| Halite     | 31.69            | 31.70          | −0.01 |
| Hematite   | 33.16            | 33.15          | +0.01 |
| Microcline | 27.51            | 27.50          | +0.01 |
| Dolomite   | 30.94            | 30.96          | −0.02 |
| Albite     | 27.92            | 27.94          | −0.02 |
| Anhydrite  | 25.42            | 25.44          | −0.02 |
| Anatase    | 25.31            | 25.28          | +0.03 |
| Siderite   | 32.02            | 32.05          | −0.03 |

Four phases appeared at first to disagree and were examined individually:

- **Barite** and **fluorite**: the calculation was right and the comparison value wrong. Barite's strongest line is 3.10 Å (28.74°), not the 3.44 Å often quoted from memory, and the full calculated list — 3.445, 3.318, 3.104, 2.835, 2.122, 2.107 Å with intensities 98, 63, 100, 53, 81, 56 — matches the powder file essentially exactly. Fluorite's 111 and 220 differ by only a few per cent in intensity and their order is not diagnostic.
- **Gypsum**: the calculation gives 4.28 Å as strongest where powder files give 7.63 Å. This is the well-known preferred-orientation artefact of gypsum powder files, caused by its perfect (010) cleavage; the source library carries an explicit preferred-orientation direction for gypsum for exactly this reason. The calculated *random-powder* pattern is correct as computed.
- **Pyrite**: positions match exactly (3.128, 2.709, 2.423, 2.212, 1.916, 1.634 Å); only the relative intensities of 220 and 311 were in question, and published values differ between sources.

### Determining the TOPAS origin-choice convention

**Reasoning.** Thirteen phases failed to import because TOPAS writes the two origin choices of centrosymmetric cubic and tetragonal groups with a trailing letter — Fd-3mS against Fd-3mZ — rather than the :1/:2 of the International Tables. Guessing the mapping

was unacceptable: the two origins differ by a translation of  $(\frac{1}{8}, \frac{1}{8}, \frac{1}{8})$ , so the wrong choice silently produces wrong intensities for spinels, magnetite, zircon and anatase.

**Method and result.** The convention was determined from the structures themselves rather than assumed. Diamond, silicon and franklinite, written S, place their 8a site at (0,0,0), which is origin choice 1. Magnetite, spinel, chromite, hercynite and magnesioferrite, written Z, place it at  $(\frac{1}{8}, \frac{1}{8}, \frac{1}{8})$ , which is origin choice 2. Zircon (I4<sub>1</sub>/amdZ) has Zr at  $(0, \frac{3}{4}, \frac{1}{8})$ , the standard choice-2 setting; anatase (I4<sub>1</sub>/amdS) has Ti at the origin, choice 1. Eight independent structures agree: **S = origin choice 1, Z = origin choice 2.** With this mapping all 205 phases import.

**Conclusion.** Structure import is verified: positions reproduce accepted values to 0.03° across twelve chemically diverse phases, the four apparent discrepancies are explained and none is an error in the program, and the origin-choice convention is established from evidence rather than assumption.

## A.4 Workflow tests on synthetic data

**Reasoning.** The reduction steps — zero error, background, diagnostics, fit — can only be tested for *correctness* against data whose answer is known in advance, which no real measurement provides. Synthetic patterns were therefore constructed from the program's own library with a known composition, a known zero error and a known background, and the workflow required to recover them.

| Test                                      | Reasoning                                                                                                                                                                                      | Result                                                    |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Zero error is recovered                   | A known offset is imposed on a synthetic pattern; the calibration must return it. Tests the peak location and the sign convention — a sign error here would shift every pattern the wrong way. | Recovered within the 0.01° step. <b>Pass.</b>             |
| Background with an accumulated 1/x term   | The 1/x term must be combinable with any main component, as specified. Tests the design matrix assembly.                                                                                       | Recovers an imposed background of that form. <b>Pass.</b> |
| All background components can be combined | Polynomial, Chebyshev, exponential and 1/x simultaneously — the worst case for conditioning of the linear solve.                                                                               | Fits without degeneracy. <b>Pass.</b>                     |
| NNLS recovers a known composition         | The central claim of the method: a synthetic mixture of known proportions must be recovered. Tests the design matrix, the solver and the share computation.                                    | Recovers the composition. <b>Pass.</b>                    |
| Kaolinite collapse is detected            | A synthetic pair with kaolinite removed from the heated mount must be diagnosed as collapse.                                                                                                   | Detected. <b>Pass.</b>                                    |
| Chlorite survives heating                 | The complement: chlorite must <i>not</i> be reported as collapsed, or every chloritic sample would be called kaolinitic.                                                                       | Survives. <b>Pass.</b>                                    |
| Smectite swelling is detected             | A synthetic pair with the expandable component shifted must be diagnosed as swelling.                                                                                                          | Detected. <b>Pass.</b>                                    |

## A.5 File readers and zero-error behaviour on real mounts

### Readers

**Reasoning.** A reader that silently misreads — an offset by one point, a wrong step, an intensity scaling — corrupts everything downstream while appearing to work. The strongest available check is that two independent encodings of the same measurement,

the instrument's native XRDML and its own exported two-column text, must agree exactly.

**Result.** All 33 XRDML files read, and for every specimen where both forms exist the two agree **bit for bit**: maximum difference 0.000 counts, identical 2 $\theta$  grids. The instrument configuration carried in the XRDML — wavelengths, goniometer radius, divergence slit, counting time — is also recovered, so a calculation can be set up to match the measurement. An initial version failed on all 33 files because intensities are held in a <counts> element rather than <intensities>; both are now accepted, as are measurements holding several appended scans.

## Zero error

**Reasoning.** The zero error was first determined with a wide search window ( $\pm 0.6^\circ$ ). Run across all eleven specimens, this produced shifts ranging to  $-0.40^\circ$ , with the three mounts of one specimen disagreeing by  $0.33^\circ$ . Mounts of the same specimen are separate slides, so they may differ by a few hundredths of a degree — but not by a third of a degree. The disagreement itself was the evidence that something was wrong.

**Diagnosis.** Inspecting the region directly showed that in the kaolinite-rich specimens there is no quartz 100 peak at all, and the routine was locking onto the kaolinite 020 band at 4.36 Å,  $0.4^\circ$  away. Quartz 100 clears every clay *basal* reflection, which is why it is the standard choice, but it does not clear the kaolinite *hk* bands.

**Result after correction** (narrow window, signal-to-noise test):

| Specimen | Air                              | Glycol | Heated    | Spread |
|----------|----------------------------------|--------|-----------|--------|
| 1        | -0.06                            | -0.06  | -0.04     | 0.02   |
| 2        | -0.06                            | -0.06  | -0.06     | 0.00   |
| 3        | -0.04                            | -0.05  | -0.05     | 0.01   |
| 4        | -0.04                            | -0.06  | -0.06     | 0.02   |
| 5        | -0.04                            | -0.04  | -0.04     | 0.00   |
| 6        | -0.05                            | -0.03  | not found | 0.02   |
| 7        | -0.02                            | +0.04  | not found | 0.06   |
| 8        | -0.06                            | +0.00  | -0.06     | 0.06   |
| 9        | -0.05                            | +0.06  | not found | 0.11   |
| 10, 11   | no calibration peak in any mount |        |           | —      |

**Conclusion.** Detected shifts now lie between  $-0.06^\circ$  and  $+0.06^\circ$ , are consistent between the mounts of a specimen to  $0.02^\circ$  in the majority of cases, and are of the magnitude expected for instrument zero and specimen displacement. The eight mounts in which quartz is genuinely absent now report that fact instead of a fabricated number. *This test is the reason the search window is narrow and the signal-to-noise test exists*; with the original settings the program would have silently displaced whole patterns by  $0.4^\circ$ .

## A.6 Development of the accompanying-mineral search

**Reasoning.** Searching a 205-phase database for the few minerals actually present is a discrimination problem, and a method that reports everything is worthless. Each candidate method was tested on a synthetic pattern of *known* content — quartz, calcite and albite in known proportions — so that false positives could be counted, and then on a real specimen.

| Method                                                     | Reasoning                                                                                                                                                                                                                                        | Result                                                                                                                                                                                                                                                                                                                    | Verdict              |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 1. Per-peak window, 2 % cell allowance                     | The obvious first approach: allow each expected line to be found anywhere within the tolerance implied by a 2 % cell deviation.                                                                                                                  | <b>190 of 205 phases scored 100 %</b> on a three-phase pattern. A 2 % allowance is a window up to 1.6° wide at high angle; on a peak-rich pattern nearly every phase finds something near every expected line.                                                                                                            | <b>Rejected.</b>     |
| 2. One coherent cell scaling + cosine similarity           | A real cell deviation stretches the whole lattice together, so all of a phase's lines must move by the same relative amount — a far stronger constraint than independent windows. Intensity agreement was added as a cosine similarity.          | Better: the three true phases ranked 2nd, 4th and 7th. But 157 phases still passed, and graphite, aluminium, copper and lead ranked at the top. A cosine similarity is dominated by whichever single line falls on the largest peak, so any phase with one reflection near a strong peak scores $\approx 1$ .             | <b>Insufficient.</b> |
| 3. Coherent scaling + proportionality + minimum line count | Replace the cosine by a test that the implied scale is consistent <i>across</i> lines, and require a phase to have enough lines in range to be judged at all.                                                                                    | Improved, but on a real specimen still ranked <b>quartz 56th of 205</b> , behind ilmenite, pyrite, cementite and faujasite — while quartz was demonstrably present, being the zero-error standard.                                                                                                                        | <b>Insufficient.</b> |
| 4. Competitive non-negative least-squares screen           | Ask the question that actually matters: does this phase explain intensity that nothing else can? Every candidate is fitted simultaneously with the clay library and with every other candidate, and ranked by the share of the pattern it takes. | On a real specimen, two phases take four to five times the share of the pattern that the next candidate down takes, and the gap between them and the rest is clean rather than graded; on a second specimen, three phases separate from the rest the same way. Adding the identified phases to the fit lowered $R_{wp}$ . | <b>Adopted.</b>      |

**Conclusion.** Position matching cannot discriminate phases in a peak-rich pattern at these levels; explained intensity can. Method 4 is the default whenever a reference library is loaded, and method 3 is retained only as a fallback for when one is not. The residual tail at 1–2 % is least-squares slack rather than mineralogy, which is why the confirmation dialog leaves it unticked.

## A.7 Crystallite thickness against measured peak width

**Reasoning.** The first fits to real specimens gave  $R_{wp}$  near 68 % while reproducing every peak *position* correctly, with the difference curve showing sharp excursions at the peaks — the signature of a width or position mismatch rather than a wrong phase. Crystallite thickness is the principal control on basal width (§2.6), so it was measured.

**Method and results.** Full widths at half maximum were measured on a real glycol-solvated mount and compared with the Scherrer prediction:

| Position (°2θ) | FWHM (°) | Assignment                   |
|----------------|----------|------------------------------|
| 8.82           | 0.117    | illite 001                   |
| 12.45          | 0.134    | kaolinite 001 / chlorite 002 |
| 17.75          | 0.134    | illite 002                   |
| 26.57          | 0.134    | quartz 101 / illite 003      |

A width of 0.117° at 8.82° implies a crystallite thickness of about 680 Å, or **68 layers**. The library in use had been built with a mean of 10 layers, which predicts a 001 width of **0.79°** — nearly seven times too broad.

**Consequence.** Crystallite thickness became a spanned library dimension (§3.3). Rebuilt with means of 5, 15 and 50 layers, the fit selected the  $N = 50$  patterns for the sharp illite, confirming the mechanism.

**An instructive negative result.** Spanning thickness alone did *not* improve the agreement:  $R_{wp}$  went from 70.8 % to 70.8 %. The fit now had patterns of the right width available and chose them, but something else was still wrong — which led directly to the next two tests. Had the thickness test been taken as sufficient, the real causes would have been missed.

## A.8 Layer spacing

**Reasoning.** With widths now correct, the residual remained concentrated at one place: the illite 001. At that point the measurement showed 7764 counts where the fit gave 1279 — the strongest clay reflection in the pattern almost entirely unexplained. A peak that is present in both but not fitted is a position problem.

**Method and results.** The basal spacing was determined from the measurement itself, using two orders so that the result is self-checking:

| Order          | Observed position (°2 $\theta$ ) | d (Å)  | Implied d(001) (Å)      |
|----------------|----------------------------------|--------|-------------------------|
| 001            | 8.884                            | 9.9459 | 9.946                   |
| 002            | 17.808                           | 4.9768 | 9.954                   |
| 003 (apparent) | 26.631                           | 3.3445 | 10.034 — <i>discard</i> |

The 001 and 002 agree on  $d(001) \approx 9.95$  Å. The third order appears to disagree, but that reflection is quartz 101 at 3.3435 Å overlapping illite 003, not illite alone — a reminder that the 3.34 Å peak of a clay pattern is rarely a clay peak only.

The library value, taken from the ICSD illite structure, is **10.022 Å**, placing the 001 at 8.816° against the observed 8.884°. The discrepancy of 0.066° is *half a peak width*. Scanning the spacing:

| d(001) assumed (Å) | 001 calculated (°) | Observed – calculated (°) |
|--------------------|--------------------|---------------------------|
| 9.900              | 8.920              | −0.038                    |
| <b>9.950</b>       | <b>8.886</b>       | <b>−0.004</b>             |
| 9.980              | 8.853              | +0.029                    |
| 10.000             | 8.836              | +0.046                    |
| 10.022 (library)   | 8.820              | +0.062                    |

**Conclusion.** The specimen's illite has  $d(001) = 9.95$  Å, not the 10.022 Å of the reference structure. This is not an error in either: illite basal spacing varies from about 9.95 to 10.10 Å with interlayer potassium content and hydration, and a single refined structure is one point in that range. Layer spacing therefore became a second spanned library dimension. On rebuilding, the fit selected  $d = 9.95$  Å of its own accord, and the illite 001 residual resolved.

## A.9 The weighting of the fit

**Reasoning.** With widths and spacings now spanned, agreement was still poor. The remaining suspect was the fit's own weighting, which had been taken from the background-subtracted intensity. That is wrong on statistical grounds: the variance of a counting measurement is set by the number of photons recorded, and subtracting a smooth background does not reduce it. Weighting by the subtracted value gives near-zero-intensity points — the flat regions between peaks, which are pure noise — the largest weights, and the peaks the smallest.

**Result.** Changing the weights to  $1/\max(\text{raw counts}, 1)$ , and weighting the solve as well as the metric:

| Weighting                                      | $R_{\text{wp}}$ | $R_{\text{p}}$ |
|------------------------------------------------|-----------------|----------------|
| By background-subtracted intensity (incorrect) | 70.8 %          | 48.2 %         |
| By raw counts (counting statistics)            | 56.0 %          | 46.4 %         |

**Conclusion.** A 15-point improvement in  $R_{\text{wp}}$  from a change that alters no physics at all, only the statistical weighting. The lesson is general: when a background-subtracted pattern is fitted, the weights must still come from the counts that were actually measured.

## A.10 Cumulative effect and the final state

**Reasoning.** The preceding tests each addressed one cause. Their cumulative effect on a single real specimen records what each was worth, and whether the final state is fit for use.

| Stage                                  | $R_{\text{wp}}$ | Change made                  |
|----------------------------------------|-----------------|------------------------------|
| Initial                                | 68.4 %          | —                            |
| + thickness spanned                    | 70.8 %          | no improvement alone (§A.7)  |
| + correct weighting                    | 56.0 %          | §A.9                         |
| + quartz in the fit                    | 53.4 %          | accompanying minerals matter |
| + quartz, albite, calcite              | 49.1 %          | —                            |
| + layer spacing spanned                | 35.7 %          | §A.8                         |
| + minerals from the competitive screen | 33.2 %          | §A.6                         |

**These figures predate the background correction of §A.11.** They are kept because each row records what one change was worth, and the ordering of the rows is unaffected; the absolute values are not comparable with the measurements below, which were made after it. A background that cut under the data inflated the observed intensity, and  $R_{\text{wp}}$  is measured against that intensity, so the earlier figures were flattered by it.

**Across all eleven specimens, after the correction.** Fitting each glycol mount with the clay library, the accompanying minerals from the competitive screen and the zero error applied,  $R_{\text{wp}}$  runs from **37.4 %** to **67.5 %** with a median of 44.8 %. Eight of the eleven fall between 37 % and 50 %. Two kaolinite-dominated separates fit at 65 % and 68 %; their 7.17 and 7.37 Å pair is not

matched by the kaolinite library and the cause is not established. Both are specimens in which quartz is absent, so their zero error had to be left unset.

**Conclusion.** Typical agreement is  $R_{wp} \approx 40\text{--}48\%$ , which is modest for a whole-pattern fit and is what a fixed library that cannot refine positions or widths continuously can be expected to give; it is poor by the standards of a Rietveld refinement, which can refine them. Two specimens of eleven are fitted badly. The plots exist so that this is visible to the user in every case, and §9 states the limitation plainly.

## A.11 The peak stripping, and what it must not remove

**Reasoning.** The background components are fitted to a peak-stripped estimate rather than to the counts, and on real mounts they follow it to a few counts. The estimate therefore is the background, and the question is what the stripping should leave alone. The answer is not a matter of taste. Where a background is convex, the mean of two symmetric neighbours is never below the point itself, so the SNIP operation - replace each point by the smaller of itself and that mean - leaves a convex background exactly unchanged. The low-angle tail of an oriented mount, air scatter and the shoulder of the direct beam, is such a background. A correct implementation returns it untouched; an implementation that removes it leaves thousands of counts of direct beam behind as signal, to be explained as diffraction by whatever pattern in the library rises at low angle.

**Method.** A background of the measured shape (1820 counts at  $3^\circ$  falling to 339 at  $30^\circ$ ) was constructed and stripped *with no peaks present at all*, at five widths. There is nothing to strip, so the estimate must return the input. Ten peaks of 400 to 7000 counts were then added and the test repeated.

| Implementation                                | Estimate at $3^\circ$ , background alone | Worst error, background alone | Worst error, with peaks |
|-----------------------------------------------|------------------------------------------|-------------------------------|-------------------------|
| As released (invented neighbours at the ends) | 546                                      | 1274 counts                   | 1270 counts             |
| Corrected (comparison narrowed at the ends)   | 1820                                     | 0 counts                      | 30 counts               |

**Result.** The released implementation removed 70 % of a background that contained no peaks whatever. The cause was at the ends of the scan, where the comparison has no neighbour on one side: the missing neighbour was replaced by the point itself, which on a falling edge puts the mean below the point, so every one of the roughly 120 passes lowered it further and the damage propagated inwards over several degrees. Padding the ends with an extrapolated trend, tried next, was no better - the padding is itself stripped and carries the data down with it - and padding by reflection inverts any peak that lies near the end. Narrowing the comparison to what the data allows, so that the window shrinks towards the ends instead of being filled in, restores the property exactly: with no peaks the estimate is the input to within a count, at every width tested.

**What it changes.** On a real air-dried mount measured from  $3.01^\circ$ , with a Chebyshev background of degree 4 plus the  $1/x$  term:

| 2θ                      | 3.01° | 4.0° | 5.0° | 8.85° | 20.85° | 34.0° |
|-------------------------|-------|------|------|-------|--------|-------|
| Measured                | 3505  | 2673 | 2369 | 3670  | 983    | 493   |
| Background, as released | 998   | 973  | 933  | 831   | 484    | 319   |
| Background, corrected   | 3294  | 2650 | 2174 | 1151  | 558    | 381   |

The released background was a nearly flat line 200 counts above the high-angle floor across the whole pattern - which is how it was reported, as a background that appeared not to respond to any control. It could not respond: the estimate it was fitted to had been flattened before the model ever saw it. Averaged over the quantified range the corrected background is 210 counts higher.

**The stripping width now means peak width.** With the tail preserved by the convexity property rather than by the choice of width, the control no longer trades the tail against the peaks. Measured over the eleven glycol mounts against the clay library alone, without accompanying minerals or zero-error correction, so the values are higher than those of §A.10 and only the trend is meaningful:

| Width (°2θ) | Overall $R_{wp}$ | 3–4.5° surviving | 8–40° $R_{wp}$ |
|-------------|------------------|------------------|----------------|
| 1           | 65.3 %           | 2.2 %            | 64.0 %         |
| 2           | 60.9 %           | 4.5 %            | 61.1 %         |
| 3           | 59.6 %           | 5.1 %            | 60.1 %         |
| <b>4</b>    | <b>59.0 %</b>    | 5.1 %            | 59.5 %         |
| 6           | 58.7 %           | 5.1 %            | 59.2 %         |
| 8           | 58.6 %           | 5.2 %            | 59.1 %         |

The share of low-angle counts surviving subtraction is now 5 % at every width from 3° upwards: the tail is background whatever is asked for, which is the property above. What the width still changes is how much of a broad reflection is treated as background, and both metrics flatten by about 4°, which is the default. Below 2° the background is lifted into the peaks and the 8–40° figure, a region containing no tail at all, is 4.5 points worse.

**The same defect in the library.** Each calculated pattern has its own diffuse continuum stripped before storage (§3.3), by the same routine, and the same end effect left that continuum standing over the first degrees - 21 % of the strongest peak at 3.5° on an I/S 0.80/0.20 pattern. Since it is a calculated pattern, the end can be moved: each is now computed on a grid reaching one stripping window beyond both ends of the stored range and trimmed afterwards, which puts the unreliable zone outside the library entirely. The residual continuum at 3.5° falls to 5 %, with the peaks unchanged to three decimal places. This matters because the measured background and the calculated continuum have to be removed consistently: if one side keeps its low-angle rise and the other does not, the fit is asked to reconcile an intensity that only one of them has.

**Conclusion.** The property to test for a background estimator is not how smooth its output looks but what it does to a background that contains no peaks. That test is now in the suite (`tests/test_background.py`), at five widths, along with the requirement that peaks are still removed and that the estimate never exceeds the measurement.

## A.12 End-to-end tests of the interface

**Reasoning.** Unit tests cannot show that the assembled program works: state may be lost between steps, a control may not be connected, a computation may be correct but never reach the screen. The complete workflow was therefore driven in a real browser, automatically, on real measurements, with the browser console monitored for errors.

**Method.** A script loaded three mounts, set the zero error on each, fitted and applied a background, loaded the phase database and reference library, ran the mineral search, confirmed the selection, ran both diagnostics, ran the fit, visited all four result plots, exported, and opened the simulator — capturing a screenshot at each step.

**Results.** The workflow completed with **no JavaScript errors**, all four plots rendering and all seven export files written. Two defects were found this way and would not have been found otherwise:

- **The confirmation dialog accepted itself.** Dash fires callbacks once for newly created components, so the dialog took whatever was ticked by default at the instant it appeared — the analyst never saw it, and three implausible minerals were silently added to the fit. The screenshot showed the selection already made. Fixed by guarding against the initial invocation.
- **Stale server state.** An early run reported results from before a change, because the development server does not reload edited code. Subsequent runs restart the server explicitly.
- **Two curves in the same colour.** The background plot drew the measured pattern in the colour of the mount and the subtracted result in a fixed green — which for the glycol-solvated mount is the same green. The two lay on top of each other, the stripped estimate was a faint grey dotted line beneath the model, and a correctly applied background was indistinguishable from none at all; it was reported as the background not being applied. The traces now have fixed, distinct colours and the estimate is drawn over the model rather than under it.
- **Tabs that did not refresh.** The interface is built once at start-up, so the background and zero-error views were still showing "not loaded" after the patterns had been read, and kept a background fitted to uncorrected angles after the zero error was set, until some control was touched. Both now take the load and zero-error status as inputs.

**Conclusion.** The assembled program works end to end on real data. The dialog defect in particular was invisible to unit testing, being a property of the interface framework rather than of any function, and was caught only by driving the interface and looking at what it displayed.

## A.12.1 A name that was never imported

**What happened.** Pressing *Load database* on the main-minerals tab reported name `'resolve_user_path'` is not defined. The function exists and is correct; the module that calls it never imported it. Python raises that only when the line runs, so the interface started, the first three tabs worked, and the fault waited on a button. The same import was missing for the two other calls in that file — loading the reference library and exporting the results — so three of the seven steps were broken in the same way and by the same omission.

**Why the existing tests did not catch it.** The unit tests exercise the library, not the interface, and the end-to-end browser run that would have caught it had last been made before those three calls were written. A test that only proves what it reaches cannot prove what it does not.

**What was added.** Two things, because they fail differently. A whole-source check for names that do not exist (`tests/test_no_undefined_names.py`, using `pyflakes`) runs over every file in a fraction of a second and finds this class of fault wherever it is, including on paths no test reaches; run against the unfixed source it named all three lines. And the browser run now covers every tab and every button through to the export, which is what establishes that the assembled program works rather than that its parts do.

**Result of that run.** Load, zero error ( $-0.06^\circ$ ), background, phase database (209 phases), mineral search, kaolinite (most of the  $7.15 \text{ \AA}$  area collapsed on heating, as it should), expandable, library (1601 patterns), fit ( $R_{wp} = 49.6 \%$ ), export (7 files) and the simulator, with no JavaScript errors.

## A.12.2 A step that looked like it did nothing

**What happened.** *Search for main minerals* was reported as doing nothing. It was doing exactly what it should, and finishing: the search takes seven to ten seconds, and Dash displays nothing at all while a callback runs. A button that shows no change for ten seconds cannot be told apart from a button that is broken, and was not.

**What was changed.** Every pane a slow step writes into is now wrapped so that a spinner appears over it while the step is computing, with the previous contents dimmed underneath rather than blanked, and a short delay before it shows so the quick steps do not flicker. That covers the mineral search, the quantification, the library load and build, and the export — the four steps measured in seconds rather than milliseconds.

| Step                                              | Time on a real measurement |
|---------------------------------------------------|----------------------------|
| Mineral search, peak matching (no library loaded) | 7.4 s                      |
| Mineral search, competitive fit (library loaded)  | 9.6 s                      |
| Quantification over 1614 patterns                 | tens of seconds            |

**And the sliders.** Each slider carried a bubble showing its value under the handle, which sat on top of the label of the control beneath it and hid part of the text. The value is already shown in the box beside every slider, so the bubbles are removed — nineteen of them.

## A.14 Weight percent from the calculated masses

**Reasoning.** The chain from a fitted scale factor to a mass has four links — the pattern is calculated for one unit of the phase, stored divided by a normalisation, fitted with a coefficient, converted back with the mass of that unit — and an error in any one of them scales a phase without looking wrong. Nothing in a real measurement can catch that, because the true answer is not known. A specimen built out of the library's own patterns, from masses chosen in advance, can.

**Method.** Known masses of four library patterns were combined into one pattern, multiplied by an arbitrary constant standing for the instrument, and fitted. The fit is exact by construction ( $R_{wp} < 10^{-6}$ ), so anything wrong in the result is the conversion rather than the fit.

| Phase                     | Mass put in | Weight % returned |
|---------------------------|-------------|-------------------|
| Illite                    | 40.0 %      | 40.00 %           |
| Kaolinite 1M              | 25.0 %      | 25.00 %           |
| Chlorite                  | 20.0 %      | 20.00 %           |
| Illite/smectite 0.80/0.20 | 15.0 %      | 15.00 %           |

**Result.** Exact to the printed precision, and unchanged when the arbitrary constant is doubled — counting twice as long does not change a weight percent. Equal masses of chlorite and kaolinite come back equal although their shares of scattered intensity differ, which is the point: the conversion uses the mass of the calculated unit and not the intensity. A library that does not record those masses reports no weight percent at all rather than a number computed from something else.

**And what it did not test.** That the relation was the right one. The specimen was built by inverting ClayQuant's own conversion, so a factor missing from both sides cancelled and the test passed. One was missing: equation (2.25) needs the volume of the calculated unit as well as its mass, and it was implemented with the mass alone.

## What caught it

A TOPAS refinement of the same oriented air-dried mount, made independently in the laboratory the measurements come from. The version of it available at the time quantifies four phases — quartz, albite, kaolinite 2*M* and sekaninaite — and fits illite and chlorite as structureless hkl\_*I*s phases with no weight at all, which is itself worth noting: an expert refinement of an oriented mount does not attempt to weigh the clays either.

The specimens are laboratory material and their compositions are not reproduced in this manual. What is given below is the *ratio* of ClayQuant's answer to the reference's, which is what a verification record needs and which says nothing about what the specimen contains.

| Phase       | Cell volume         | ClayQuant / reference, without <i>V</i> | ClayQuant / reference, with <i>V</i> |
|-------------|---------------------|-----------------------------------------|--------------------------------------|
| Quartz      | 113 Å <sup>3</sup>  | 1.81                                    | <b>1.10</b>                          |
| Albite      | 664 Å <sup>3</sup>  | 0.39                                    | <b>1.38</b>                          |
| Kaolinite   | 657 Å <sup>3</sup>  | 0.20                                    | 0.35                                 |
| Sekaninaite | 1551 Å <sup>3</sup> | 0.40                                    | 3.2                                  |

Quartz has the smallest cell of the four and was over-weighted against albite by 4.7 times, which is the ratio of their cell volumes, 5.9, to within the other differences between the two analyses. With the volume included, quartz and albite agree with the refinement to 10 % and 38 % where they were out by factors of two and five — and the same correction cuts the quartz of the glycol-solvated mount of that specimen by two thirds, leaving the clay minerals carrying most of the mass, which is what a clay separate should look like.

Two differences remained and were not explained by this: kaolinite at a third of the refined value and sekaninaite at three times it. A later version of the refinement accounts for both, and is described next.

## The refinement that weighs every phase

The refinement above was superseded by one of the same measurement in which illite and chlorite are modelled as structures rather than as hkl\_*I*s peak sets, every phase carries a refined March–Dollase parameter instead of a fixed one, and all seven phases are weighed to a sum of 100 %. It is the better reference in every respect, and it changes the earlier comparison as much as it confirms it. Kaolinite and albite both fall by more than half once their texture is refined rather than assumed absent, which is where two of the disagreements above went; and rutile — which ClayQuant had been finding, and which was suspected of being a false positive — is in it as a phase of a few per cent.

| Phase                  | ClayQuant / reference | Reference $r$ , relative to ClayQuant's                               |
|------------------------|-----------------------|-----------------------------------------------------------------------|
| Quartz                 | 0.61                  | both fixed at 1                                                       |
| Chlorite (clinochlore) | <b>0.88</b>           | within a factor of 1.8                                                |
| Kaolinite              | 0.50                  | slightly more textured                                                |
| Albite                 | 3.20                  | textured about $\theta^2 \theta$ , which ClayQuant cannot express     |
| Rutile                 | <b>0.89</b>           | textured about $1 \ 1 \ \theta$ , which ClayQuant cannot express      |
| Illite                 | 7.8                   | far more textured than the fit chooses                                |
| Sekaninaite            | 11                    | textured about $2 \ \theta \ \theta$ , which ClayQuant cannot express |

ClayQuant's figures are renormalised over these seven phases; the competitive screen also admitted six minerals the refinement does not use — microcline, anorthite, tridymite, thaumasite, a calcium phosphate and pseudoboleite — carrying a further quarter of its mass, and the first three of those sit on feldspar and silica lines. ClayQuant's  $R_{wp}$  for this mount is 36 %, against the refinement's 6.7 %: a fixed library fitted linearly is not a Rietveld refinement and does not pretend to be one.

**What this settles.** Chlorite at 0.88 and rutile at 0.89 are the first phases to agree with an independent analysis to better than the uncertainty of the preparation, and chlorite could not be compared at all before, since the earlier refinement gave it no weight. Chlorite is counted here as the chlorite and chlorite/smectite entries together, which is what a single clinochlore phase in the refinement corresponds to. Rutile is real.

**What it does not.** Illite is nearly eight times the refined value and albite three times. Both have an identifiable cause and they are different causes.

Illite is the texture parameter, and the refinement measures it where ClayQuant cannot: it settles on a far stronger texture than the fit here chooses for its illite and illite/smectite entries. Because a basal series has the same shape at every  $r$  (§2.9), that difference is invisible in the residual and enters the mass as  $r^3$ . Holding ClayQuant's illite at the refined value and leaving everything else alone takes the ratio from 7.8 to 3.4 — most of the discrepancy, and exactly the quantity the following section says the measurement cannot determine.

Albite is a limitation of the model rather than of the fit. Its refined texture is about  $\theta^2 \theta$ , and sekaninaite's about  $2 \ \theta \ \theta$  — axes that are not  $c^*$ . ClayQuant applies March–Dollase about  $c^*$  only, which is right for a platy clay and wrong for a feldspar lying on a cleavage that is not basal, so it has no way to express either texture and assumes both minerals are randomly oriented. Offering the accompanying minerals at several values of  $r$  about  $c^*$  rather than at 1 alone brings albite from 3.20 down to 2.14 and leaves the rest where they were, which is about as much as an axis-restricted correction can do. A general orientation axis per phase is the fix, and it is not implemented.

Expressed as a mass ratio to quartz, which does not move when another phase does, the seven come out at 1.00 for quartz, 0.82 for kaolinite, 1.45 for chlorite, 1.46 for rutile, 5.2 for albite, 12.8 for illite and 18 for sekaninaite. Kaolinite, which looked like a factor-of-two problem in the renormalised table, is the second-best phase on this measure; what moves it is quartz, which ClayQuant under-weights by a third.

**What no refinement of this mount can supply.** It is the *air-dried* mount, and it contains no expandable phase: no smectite, no montmorillonite, no illite/smectite. That is not an omission but a consequence of the preparation — in an air-dried specimen an expandable clay sits near 14–15 Å, on top of chlorite 001 and vermiculite, and cannot be separated from them. That is what the

glycol-solvated mount is for (§2.1). The reference therefore constrains six crystalline phases and the illite, and says nothing about the expandable clay, which is the quantity the three-mount method exists to measure.

## How well the fit determines the texture parameter

The illite discrepancy above is mostly the texture parameter, so it is worth measuring what the parameter is worth. For the kaolinite entries of the library, from  $r = 0.1$  to  $r = 1$ :

| $r$ | Similarity of the stored pattern to $r = 1$ | Mass per unit coefficient |
|-----|---------------------------------------------|---------------------------|
| 0.1 | 0.842                                       | $3.8 \times 10^{-5}$      |
| 0.5 | 0.861                                       | $4.8 \times 10^{-3}$      |
| 1.0 | 1.000                                       | $3.8 \times 10^{-2}$      |

A tenfold change in  $r$  changes the shape of the pattern by 16 % and the mass it implies by a factor of a thousand. The shape it changes is the ratio of  $hkl$  to  $00l$  intensity, which in a strongly oriented mount is the weakest thing in the pattern. The weight percent of a basal-dominated phase is therefore not merely uncalibrated: it rests on a quantity the measurement barely determines.

That has a consequence for the calibration of §2.13. A factor  $k_p$  fitted while  $r$  is also free would be absorbing a parameter the data cannot fix. The calibration should instead *fix*  $r$  per phase, determined from the specimens of known composition, so that afterwards the fit chooses only how much of each phase is present.

The refinement bears this out from the other side. It determines  $r$  for each phase because it fits the  $hkl$  reflections as well as the basal ones, in a pattern where those reflections are weak but present; ClayQuant's clay library is basal-only by construction, so nothing in it constrains  $r$  at all. Where the two nearly agree on  $r$  — chlorite — they agree on the weight to 12 %; where they do not, they disagree by the cube of the ratio.

## The importer now checks itself against TOPAS

A structure block written by TOPAS states two numbers ClayQuant does not need, because it works both out from the sites and the cell: `cell_mass` and `cell_volume`. That is what makes them worth reading. They are the same two quantities stated independently, by the program the library was written for, so a block that has lost atoms disagrees with itself — and nothing else in the import notices, since a phase with some of its sites is still a phase that reads.

The importer now compares them and reports every phase that disagrees by more than 0.5 %. Over the 209-phase laboratory library exactly one does, and it is the phase that had been added by hand:

phases that were imported but do not match what TOPAS says they weigh:  
Sekaninaite: cell mass works out to 1696.146 from the sites and cell given,  
but the block states 2592.043 (65 % of it); sites are probably missing

Five oxygen sites — O2 to O6 — are absent from that item; they are present in the refinement's own output for the same phase. The cell volume agrees exactly, which localises the fault to the site list rather than the cell. Restoring the five sites brings the computed mass to 2592.090 against the stated 2592.043. The phase is still written to the database, because whether an incomplete structure is usable is the user's judgement and not the importer's; it is only no longer silent. Sekaninaite is a trace phase here, so little followed from it, but the same fault in a major phase would have gone the same way.

**Conclusion.** A test built by inverting the code it tests will agree with the code and not with nature. Both faults in this section were found by comparison with something that did not come from ClayQuant: the missing cell volume by a refinement of the same measurement, the missing oxygen sites by the refinement's own statement of what the structure weighs. The synthetic test remains, rebuilt from the powder relation rather than from ClayQuant's conversion, alongside a check that the two calculation routes for one phase — as a discrete cell and as the host layer of a stack — weigh a specimen alike.

## A.15 A weighed standard, and a random mount ClayQuant cannot fit

**Reasoning.** The oriented mounts confound two things: whether the conversion from a scale factor to a mass is right, and whether each phase's texture is modelled. A random powder mount removes the second — there is no texture and every phase is fitted at  $r = 1$  — so what remains is the conversion alone. And a random mount can carry a weighed standard, which makes the reference absolute rather than relative. The laboratory's bulk analyses are exactly this: pressed random powders with 20 % of corundum by weight added before mounting, refined in TOPAS.

**What the spike says about the reference analyses.** The refinements record no trace of the spike — the weight of corundum added is knowledge held outside the files — so their weight percent is relative, normalised to 100 over the phases that were weighed. Applying (2.26) makes them absolute, and the difference from 100 is what each refinement does not account for: its illite and chlorite, which it fits as hkl\_Is peak sets carrying no weight, together with any amorphous content. Sixteen refinements, eight specimens each prepared twice:

| Quantity, over eight specimens                         | Lowest | Highest | Largest difference between the two preparations of one specimen |
|--------------------------------------------------------|--------|---------|-----------------------------------------------------------------|
| Corundum, refined against 20 % added                   | 21.0 % | 44.5 %  | 2.3 %                                                           |
| Scale factor implied, $W_{\text{std}}/W'_{\text{std}}$ | 0.45   | 0.95    | 0.03                                                            |
| Unaccounted, of the spiked powder                      | 4.9 %  | 55.0 %  | 2.8 %                                                           |
| Unaccounted, of the specimen as sampled                | 6.2 %  | 68.8 %  | 3.5 %                                                           |

The per-specimen compositions are laboratory data and are not reproduced here; what the verification argument needs is the spread and the reproducibility. The two preparations of a specimen agree to within 0.6 to 2.8 percentage points on the unaccounted fraction, which is the reproducibility of the whole chain — splitting, spiking, pressing, measuring and refining — and is the figure against which any disagreement should be read. The unaccounted fraction ranges from 5 % to 55 % of the spiked powder, so it is not a correction but the largest single number in several of the analyses.

Note the two bases. A weight percent from (2.26) is a fraction of the powder the diffractometer saw, which is one part standard to four parts specimen; the specimen as it was sampled is the remaining 80 %, so every phase in it, and the unaccounted fraction with them, is larger by 1.25. That is the basis an amorphous content is normally quoted on, and `Quantification.original_basis` returns it, with the standard removed since it is not part of the specimen.

### The test that could not be run

ClayQuant was fitted to the same random mount, over the phases the refinement uses so that phase selection was not part of what was being tested, every phase at  $r = 1$ , with the corundum declared as the internal standard. The result is not a usable check, and the reason is worth recording rather than the numbers:

| Window   | 8–16° | 16–24° | 24–32° | 32–40° | 40–48° | 48–56° | 56–64° | 64–72° |
|----------|-------|--------|--------|--------|--------|--------|--------|--------|
| $R_{wp}$ | 94 %  | 60 %   | 45 %   | 42 %   | 50 %   | 47 %   | 40 %   | 54 %   |

Overall  $R_{wp}$  is 43 % against the refinement's 7.6 %, at the best setting found, and the misfit is spread across the whole range rather than concentrated anywhere.

What makes the result unusable is not its size but its instability. Shifting the pattern by the zero error the refinement reports, 0.035°, and then by twice that in each direction, with everything else held fixed:

| Shift applied                  | −0.070° | −0.035° | 0    | +0.035° | +0.070° |
|--------------------------------|---------|---------|------|---------|---------|
| $R_{wp}$                       | 67 %    | 51 %    | 43 % | 54 %    | 61 %    |
| Calcite, against the reference | 0.45    | 0.54    | 0.69 | 1.05    | 1.32    |
| Quartz                         | 0.63    | 0.82    | 0.84 | 0.69    | 0.56    |

A tenth of a degree of alignment moves calcite by a factor of three. Nothing about the mass conversion can be read through a sensitivity that large, and the sensitivity has a cause: the reference refinement models a specimen displacement of 0.061 mm, which displaces each reflection by an amount varying as  $\cos\theta$ , and ClayQuant has no such model. No single constant zero error can align peaks across 8–70° in its place, so the fit trades one end of the range against the other and the phase weights follow. Over the 3–40° of an oriented clay mount the same displacement is nearly a constant offset and the zero-error correction absorbs it; over a bulk scan it is not. The other absences compound it — one fixed pseudo-Voigt where the reference convolves an axial-divergence model and a slit width, no sample-transparency correction, and no per-phase cell, size or strain — and 70 % of the weighted misfit lies between the peaks rather than on them, which is the background left over when peaks sit in the wrong places.

The background model is *not* the cause: Chebyshev degree 4, 6 and 8, with and without an inverse term, change  $R_{wp}$  by 0.2 percentage points and calcite by 0.01. That was worth checking, because it was the first explanation that suggested itself and it was wrong.

**Conclusion, and what it does not license.** The internal-standard arithmetic is implemented and tested, and the reference analyses above are absolute because of it. But the corundum-spiked mounts do *not* supply the calibration of §2.13, for two independent reasons, either of which alone would be enough:

- They are a different material. The spiked mounts are bulk rock powders; the oriented mounts are the fine separate of the same rock. For the specimen of §A.14 both refinements exist, and they settle it directly: calcite and dolomite-ankerite are both major constituents of the bulk, and *neither appears in the oriented mount at all*, while kaolinite and rutile appear in the oriented mount and in no form in the bulk. The separation removes the carbonate and concentrates the fine, dense accessories, so the composition of one preparation is not the composition of the other, and nothing in either measurement says what fraction of the rock the separate is.
- The reference refinements leave illite and chlorite unweighed. Even for the bulk rock they give clay plus amorphous as one number — for the specimen of §A.14, a substantial one, reproducible between its two preparations to 2 percentage points — so they do not state the clay composition that  $k_p$  would be fitted against.

What would supply it is the pairing described in §2.13: the *same separate* mounted twice, once as a spiked random powder and once oriented. The random mount fixes the composition without texture; the oriented fit of that known composition then yields  $k_p$  directly. For ClayQuant to fit the random half of such a pair, the shortcomings listed above would have to be addressed first.

## A.16 Structures as refined, rather than as published

**Reasoning.** Every structure the library calculates from is of somebody else's specimen. For quartz that hardly matters; for a clay it may matter a great deal, because the octahedral sheets accept iron for magnesium and for aluminium in proportions that differ from one deposit to the next, iron scatters about twice as strongly as the magnesium it replaces, and the substitution sits at a definite height in the layer — so it changes the basal structure factors, and therefore every basal intensity ratio the quantification rests on. Those occupancies are refinable against a measured pattern. Where a refinement of the same specimen exists, it has already measured them.

**What the reference refinement of §A.14 actually refined.** Its cells, and seven occupancy parameters across three phases: the two octahedral sites of the illite, four octahedral and two tetrahedral sites of the chlorite, and the Mg/Fe site of the cordierite. No atomic position and no displacement parameter was refined for any phase, so positions and  $B$  come from the published structures either way. The chlorite refines to substantially more iron than the published clinocllore carries, and unevenly between its four octahedral sites; the illite takes iron on one of its two sites and none on the other. The values are this specimen's and are not reproduced here.

Two of the seven parameters are worth distrusting rather than using blindly. Both tetrahedral Si/Al parameters of the chlorite refined to a bound, leaving an all-aluminium tetrahedral sheet, which is not a chlorite; silicon and aluminium differ by one electron, so the data cannot separate them and the parameter drifts. It costs little — about 0.7 % of the cell mass and nothing measurable in the pattern — but a parameter at its limit is a parameter the measurement did not determine, and the same is true of the specimen displacement in the same file.

**Method.** TOPAS writes a refined value back in the syntax it read, so the refinement's output is a structure library and the same importer serves. The clay structures were taken from it, the pattern library recalculated, and the fit of §A.14 repeated. To separate the refined occupancies from the refined cells, a third library was built with the refined sites placed on the published cells.

| Structures used                               | $R_{wp}$      |
|-----------------------------------------------|---------------|
| Published throughout                          | 35.9 %        |
| Clays: refined occupancies, published cells   | 31.7 %        |
| Clays: refined occupancies and cells          | <b>31.2 %</b> |
| ... and the accompanying minerals refined too | 34.9 %        |

**Result.** Taking the clay structures from the refinement improves  $R_{wp}$  by 4.7 percentage points, the largest single improvement recorded in this appendix after the fit weighting of §A.9, and the occupancies account for 4.2 of the 4.7. The agreement ratios of §A.14 barely move: chlorite from 0.94 to 1.08, illite from 7.5 to 7.3, quartz unchanged at 0.6. That combination is coherent rather than disappointing.  $R_{wp}$  measures the shape of the calculated pattern, which the occupancies fix; the weights are governed by the  $r$ -degeneracy and by texture axes the model cannot express (§A.14), which a structure cannot touch.

The effect on the calculated basal series, as the cosine similarity of the normalised pattern to the one from the published structure, locates it:

| Phase        | Refined cell only | Refined sites only | Both  |
|--------------|-------------------|--------------------|-------|
| Chlorite     | 0.826             | 0.961              | 0.810 |
| Illite       | 0.953             | 0.989              | 0.947 |
| Kaolinite 2M | 0.993             | 1.000              | 0.993 |

Chlorite moves most, and its 003 — the reflection an octahedral iron content is refined against — falls by a factor of about 2.6 relative to 002. Kaolinite barely moves, which is expected: its occupancies are all 1 and only its cell changed. Note that the cell changes the isolated pattern more than the sites do, while the sites improve the *fit* more. That is not a contradiction: a changed cell mostly moves the basal peaks, and the library spans layer spacing (§A.8), so the fit absorbs a spacing change by choosing a different spanned entry. A changed intensity ratio it cannot absorb at all.

### Why not the accompanying minerals as well

Because it is measurably worse, by 3 points of  $R_{wp}$ , and the phase responsible is quartz: taking quartz's refined cell alone costs 3.0 points, albite's alone 1.3, and rutile's and cordierite's nothing. Quartz is a special case with an identifiable mechanism. It is also the zero-error standard (§5.3), and the standard is a fixed  $d(100)$  constant rather than the library's quartz: align the pattern to that constant and then fit a quartz whose refined cell puts its own 100 elsewhere, and the standard and the structure are out of step by  $0.027^\circ$ . Making the standard consistent with the refined cell recovers 1.2 of the 3.0 points. The remaining 1.8 is not explained; the refined quartz cell is simply a worse description of these data than the published one, in a program that cannot re-refine it.

A first hypothesis was that the refined cells carried the refinement's own alignment corrections — it fits a zero error and a specimen displacement, the displacement at its limit, where ClayQuant fits only a constant zero error. It was wrong: removing ClayQuant's zero-error correction makes every case worse, refined and published alike. Recorded because it was plausible and is the sort of explanation that is easier to assert than to test.

**Conclusion.** Refined clay structures are used where a refinement supplies them, and are the recommended input; the accompanying minerals stay on the library structures unless a library structure is demonstrably wrong, as the incomplete cordierite of §A.14 was. What a refinement cannot supply is the commonest case: an oriented mount refined with its clays as hkl\_Is peak sets has no clay structure in it, and the import reports that rather than succeeding emptily. 21 automated tests cover the reading, the name mapping, the substitution inside the loader, and the restoration of the published structures afterwards.

### A.17 Against measurements of the pure minerals

**Reasoning.** Every comparison up to here has been against an analysis of a mixture, where a disagreement can lie in the calculation, in the phase selection, in the texture of one phase among several, or in the reference itself. A measurement of a single pure mineral removes all of that at once. And one quantity in such a pattern is free of the two things a pure mount does not reveal — how much material is in the beam, and how well oriented it is: the *ratio* of one basal intensity to another. Preferred orientation multiplies a whole basal series by one factor (§2.9), so it divides out completely, and so does the amount of material. What is left in a basal ratio is the structure factors, the Lorentz–polarization factor, the beam-overflow correction and the crystallite thickness distribution.

**Method.** Oriented air-dried mounts of separate pure clay minerals, measured on one instrument under one setting within fourteen hours: 240 mm radius, graded incident mirror, 0.03 and 0.02 rad Soller slits, fixed 0.5° divergence and anti-scatter slits, 14 mm mask, 0.0167° steps, 19.7 s per step, no attenuator. Each 00*l* was integrated over ±1.0° of where the layer spacing puts it, after peak-stripped background subtraction, and divided by another order of the same pattern. The calculation used the published structures and the same instrument description, at the crystallite thickness that best matches the measured peak widths.

**The integration window is not a detail, and getting it wrong produced a confident and wrong conclusion.** A clay basal reflection has a full width at half maximum near 0.1–0.2°, so ±0.35° looks more than ample. It is not. Stacking disorder gives these reflections a long diffuse tail, and on one of the chlorites below a ±0.35° window held only 36–72 % of each order's area — and a different fraction of each order, which is precisely what a ratio of orders is sensitive to. At ±1.0° the same pattern gives 85–93 %. The 005 ratio of that chlorite reads 0.77 at ±0.35°, 1.22 at ±0.6° and 1.34 at ±1.0°. An earlier version of this section reported the ±0.35° figures and drew a conclusion from them that the corrected figures do not support; the retraction is below.

| 00 <i>l</i> | Kaolinite   | Chlorite A | Chlorite B |
|-------------|-------------|------------|------------|
| 001         | 1.00        | 0.94       | 0.99       |
| 002         | <b>1.05</b> | 1.00       | 1.00       |
| 003         | 0.92        | 0.94       | 1.18       |
| 004         | —           | 1.34       | 1.32       |
| 005         | —           | 1.34       | 1.18       |

Observed divided by calculated, each column normalised on the order shown as 1.00 — 001 for kaolinite, 002 for the chlorites, 002 being the strongest chlorite basal. Integrated over ±1.0°. The 006 of both chlorites is omitted: the calculated 006 falls close to a node of the structure factor, so its ratio is set by noise rather than by the model.

**Result.** Kaolinite's 002/001 comes back at **1.05**, and it is stable against the window: 1.02 at ±0.35°, 1.05 at ±1.0°, 1.08 at ±1.5°. That is the strongest single piece of evidence in this appendix that the chain of §2 is right — an absolute basal intensity ratio of a real mineral, reproduced to about 5 % with nothing fitted. Its 003 is 0.92, having read 0.65 at the narrow window.

Both chlorites come out about 1.3 on 004 and 1.2–1.3 on 005: the calculation *under*-produces the higher orders, by a similar amount in both specimens. Their 003 differs (0.94 against 1.18), which is the one place they are not alike.

### A retraction

Read at ±0.35° and at a crystallite thickness fixed at 25 layers rather than matched to the observed widths, the same two chlorites appeared to depart from the published clinochlore *in opposite directions* — 0.62 against 0.83 on 003, 0.61 against 0.81 on 005. That looked like strong evidence that the two specimens differ in octahedral iron, that no single structure can serve both, and that a per-specimen composition was needed. Corrected, the two chlorites agree with each other to within 0.02 on 004 and differ from the published structure in the *same* direction on every order but 003. The pattern is a common shortfall in the calculated higher orders, not a composition difference between specimens, so this data does not call for a per-specimen chlorite composition; the published structure remains the better-supported default.

Fitting the iron content anyway bears that out rather than contradicting it. On these two patterns the two-parameter fit of §4.4 reduces the residual by only 1.5 to 1.7 times, drives one specimen to essentially zero octahedral iron — which for a mineral sold as an iron-rich chlorite is not credible — and settles against the lower bound of the crystallite thickness grid. A parameter at its bound is a parameter the measurement has not determined, and three signs of a model straining to fit are worth more than the improvement in the number being minimised. The mechanism is available and is verified against patterns calculated from known compositions, which it recovers; what these measurements do not show is a need for it.

### What the chlorite shortfall is not

The chlorites need their 004 raised by about a third against their 002, and their 005 by a fifth to a third. The obvious suspect is a displacement parameter:  $\exp(-Bq^2)$  falls with angle, so a  $B$  that is too large suppresses the high orders. The exponential is weak here — at 005 of a 14.28 Å spacing  $q^2$  is  $0.031\text{ Å}^{-2}$ , so  $B = 0.65$  costs 2 % of an amplitude — but  $F(00l)$  is a sum of terms with opposing signs and the high orders are small differences of large numbers, so the sensitivity does not follow from the size of the exponential and has to be computed. Computed, with the 004/002 ratio the quantity to move from 0.62 to 0.83:

| Change made to the calculation                                  | 004/002 reached | Verdict                                            |
|-----------------------------------------------------------------|-----------------|----------------------------------------------------|
| Octahedral $B$ from 0 to $6\text{ Å}^2$                         | 0.63 → 0.57     | Ruled out; and the wrong way round                 |
| Every $B$ in the structure scaled 0 to $6\times$                | 0.64 → 0.55     | Ruled out                                          |
| Hydroxyl $B$ from 0 to $8\text{ Å}^2$                           | 0.62 → 0.63     | Ruled out                                          |
| $\text{Fe}^{2+}$ in place of $\text{Fe}^{3+}$                   | 0.62            | Ruled out                                          |
| Interlayer hydroxide sheet moved $\pm 0.3\text{ Å}$ along $c^*$ | 0.48 → 0.57     | Ruled out, and it destroys 001                     |
| Crystallite thickness, 4 to 40 layers                           | 0.63 → 0.62     | Ruled out (it is the 001 control, not the 004 one) |
| Single-crystal Lorentz factor instead of the powder one         | 1.25            | Overshoots, halves 001, and contradicts §A.2       |
| Beam-overflow specimen length 25 → 15 mm                        | 0.62 → 0.80     | Reaches it — see below                             |

So the displacement parameters cannot do it. The whole range of 004/002 accessible to any  $B$ , at any value from zero to six times the published one, is 0.55 to 0.64, and the direction a shortfall needs is exhausted within 2 % of the published value. Neither can the iron valence, the hydroxide sheet's height, or the crystallite thickness.

The one change that reaches 0.83 is not structural at all: the specimen length in the beam-overflow correction (§2.10), which had been assumed at 20 mm rather than measured. At 15 mm the chlorites' 001, 004 and 005 all land within a few per cent. But the cross-check refutes it. The three mounts were prepared the same way, so one length must serve all three, and it does not:

| Specimen length assumed | 12 mm | 15 mm | 16 mm        | 18 mm | 20 mm        | 25 mm        |
|-------------------------|-------|-------|--------------|-------|--------------|--------------|
| Kaolinite               | 0.38  | 0.22  | 0.18         | 0.09  | <b>0.065</b> | <b>0.065</b> |
| Chlorite A              | 0.22  | 0.14  | <b>0.14</b>  | 0.19  | 0.25         | 0.26         |
| Chlorite B              | 0.19  | 0.06  | <b>0.050</b> | 0.14  | 0.21         | 0.22         |

Root mean square of obs/calc – 1 over the orders other than the reference, at the crystallite thickness that best fits each row.

Kaolinite wants 20 mm or more and the chlorites want 15–16, and the disagreement is not marginal: at 16 mm kaolinite's residual triples. The reason the two disagree is worth noting, because it says what the parameter is doing. Kaolinite's reference order and the chlorites' both sit at  $12.4^\circ$ , and 20 mm puts full illumination at  $12.0^\circ$  — so at 20 mm and above the correction has stopped acting on any of these ratios at all, which is why kaolinite's residual is flat there. Shortening the specimen brings the reference order back into the overflow region and lifts the higher orders against it. That lifts the chlorites onto their measured values and lifts kaolinite off its own. A parameter that can absorb one discrepancy while creating another is not the cause of either.

**So it remains open.** What has not been tested: the individual oxygen and hydrogen heights within the chlorite layer, as against the rigid shift of the whole hydroxide sheet tried above; interstratified smectite or vermiculite layers in a natural chlorite, which would add intensity at low angle and redistribute the orders; and a different chlorite structure altogether. The integration window works *against* the observed excess rather than explaining it — the measured peaks are broader than the calculated ones, so a finite window discards more of the observation than of the calculation, and the true excess is if anything larger than reported.

**What this does not cover.** Only kaolinite and the two chlorites are single-phase enough to use. Of the other mounts measured, an illite's strongest reflection is quartz rather than illite, two smectites and a saponite carry several per cent of quartz, a bentonite carries opal-CT, and a synthetic hectorite is close to X-ray amorphous. None of those can serve as a reference for a basal ratio until the accompanying phase is either removed or modelled. All the mounts are also air-dried, while the quantitative fit is made on the glycol-solvated mount (§2.1), so nothing here bears on the expandable clays at all.

**And the absolute scale.** A basal *ratio* needs no standard, which is why these figures can be given at all. The absolute scale — the  $k_p$  of §2.13 — needs one, measured through the same optics. A corundum mount measured alongside these was run with a  $0.25^\circ$  divergence slit where the clays had  $0.5^\circ$ , and with a fixed slit the ratio between two such settings is not a constant: where the beam overflows the specimen both slits illuminate the same area and the ratio is 1, where neither overflows it is 2, and the crossover falls between  $2\theta = 6^\circ$  and  $12^\circ$  at a place that depends on the specimen length. That is exactly where the diagnostic basal reflections are — glycol-smectite 001 at  $5.2^\circ$ , chlorite 001 at  $6.2^\circ$ , illite 001 at  $8.8^\circ$ , kaolinite 001 at  $12.3^\circ$ . A standard measured through the clays' own optics avoids putting a modelled factor of up to two across the very peaks the calibration is for. That standard has since been measured, on these mounts' own settings, and §A.18 reports what it gives — including a check on the orientation correction that bears directly on the shortfall left open above.

## A.18 A standard through the clays' own optics

**Reasoning.** Everything in §A.14 through §A.17 is either a ratio or a relative proportion. The one number none of them reaches is the absolute one: how much mass a fitted coefficient stands for in an oriented mount, the  $k_p$  of §2.13. An internal standard cannot supply it, because a spike does not orient as the clays do (§A.15). An external standard can, because it is never asked to: it is measured on its own, and only the instrument is shared. And on a mount of a *pure* mineral the answer needs no independent analysis at all — the weight fraction is the purity, so (2.28) has a known answer and  $k_p$  is the amount by which it misses.

**Method.** A pressed corundum mount was measured under the settings of the pure clay mounts of §A.17, and the agreement is exact rather than approximate — which matters, because the previous corundum available had a  $0.25^\circ$  divergence slit against the clays'  $0.5^\circ$ , and with a fixed slit that is a factor of up to two placed across the diagnostic low-angle peaks:

| Setting                        | Corundum            | The four clay mounts |
|--------------------------------|---------------------|----------------------|
| 2θ range, step count           | 3.008–39.990°, 2214 | 3.008–39.990°, 2214  |
| Counting time per step         | 19.685 s            | 19.685 s             |
| Divergence / anti-scatter slit | 0.5° / 0.5°         | 0.5° / 0.5°          |
| Mask                           | 14 mm               | 14 mm                |
| Tube                           | 45 kV, 40 mA        | 45 kV, 40 mA         |

All five measured within ten days on the same configuration. ExternalStandard.comparable\_with compares these fields and reports nothing to flag; on the earlier corundum it named the slit difference.

The mass attenuation coefficients that (2.27) and (2.28) need were computed from each structure's mass fractions and checked against published values for the compounds:

| Phase     | $\mu/\rho$ , cm <sup>2</sup> /g | Published |
|-----------|---------------------------------|-----------|
| Corundum  | 31.14                           | 31.1      |
| Quartz    | 34.45                           | 34.4      |
| Calcite   | 74.94                           | 74.8      |
| Kaolinite | 29.77                           | ≈ 30      |
| Illite    | 42.12                           | —         |
| Chlorite  | 39.36                           | —         |

At Cu Kα, from Hubbell & Seltzer (1995). The elemental table stops where its authors' does: an element it does not hold raises rather than being given a plausible value, and the discontinuity across the Cu absorption edge is preserved — iron is 308 cm<sup>2</sup>/g and nickel, two places further on, is 49, which is not a typographical error but the reason an iron-rich clay and a corundum cannot be compared without this correction.

**Result.** Each clay mount was fitted with three-dimensional powder patterns of its own mineral spanning the orientation parameter, so that the mount's own hkl-to-basal ratio determines *r* rather than leaving it free (§A.14). The weight fraction each mount was credited with is its own purity, from the accompanying phases in the same fit.

| Mount      | Fitted as | <i>W</i> | $\mu/\rho$ | <i>R</i> <sub>wp</sub> | <i>r</i> | <i>r</i> <sup>−3</sup> | <i>k<sub>p</sub></i> |
|------------|-----------|----------|------------|------------------------|----------|------------------------|----------------------|
| Kaolinite  | kaolinite | 0.86     | 29.77      | 71 %                   | 0.16     | 244                    | <b>0.245</b>         |
| Chlorite A | chlorite  | 0.97     | 39.36      | 94 %                   | 1.00     | 1.0                    | <b>0.216</b>         |
| Chlorite B | chlorite  | 1.00     | 39.36      | 94 %                   | 0.68     | 3.2                    | <b>0.281</b>         |
| Illite     | illite    | 0.81     | 42.12      | 86 %                   | 0.59     | 4.9                    | <b>0.247</b>         |

Corundum standard: *R*<sub>wp</sub> 41 % over the whole range, 33 % over 20–39°, fitted *r* = 0.99 as a pressed powder should be. *K* = 2.28 × 10<sup>4</sup> in the units of (2.27).

Two things in that table are worth separating. The first is the spread: **0.216 to 0.281, mean 0.247, a relative standard deviation of 11 % and a ratio of extremes of 1.30**, across four mounts of three chemically different minerals. Texture, microabsorption and structural error are the three things  $k_p$  lumps together, and all three differ between these minerals — so a factor they share to within 13 % is not mostly any of them. It is a property of the comparison between a settled clay film and a pressed powder, which is what these four mounts have in common and the corundum does not.

The second is a cross-check that falls out of it, and it is the stronger result. The orientation correction  $r^{-3}$  (§2.9) is the largest single factor in an oriented-mount mass, and across these four mounts it ranges over **a factor of 244** — from a near-perfectly plated kaolinite at  $r = 0.16$  to a chlorite that behaves as a random powder at  $r = 1.00$ . The mass inferred for each depends directly on it. If the  $r^{-3}$  scaling of mass were wrong, four specimens spanning that range could not land on one factor; they land within 1.30. This is the first evidence in the manual that the orientation correction is right in magnitude and not merely in shape, and it comes from data rather than from the model.

## What the missing factor of four is not

The calculation over-produces by  $1/0.247 \approx 4$  per unit mass in these mounts. The first suspect is specimen thickness, because (2.28) is derived for a specimen thicker than the beam penetrates and a settled clay film need not be. It is refuted twice over.

Refuted by arithmetic: at the packing density of a settled film the penetration depth  $\sin\theta/2\mu$  is 3.5  $\mu\text{m}$  at 6° and 20  $\mu\text{m}$  at 35° for illite, so a deposit of 50  $\mu\text{m}$  returns 100 % of the infinite-thickness intensity below 12° and 88 % at 35°, and 100  $\mu\text{m}$  returns 99 % everywhere. The thinness needed to lose three quarters of the intensity is 1.3–2.2  $\mu\text{m}$  depending on packing — below a single kaolinite booklet's worth of stacking, and far below any mount that survives handling.

Refuted by the data, which is the part that settles it: absorption in a thin film is *angle-dependent*, and in the direction opposite to what is observed. A film thin enough to give 0.25 at the illite 001 returns 0.071 at 35°, so the higher basal orders would come in a factor of 3.5 *low* against the first. §A.17 measures the higher orders of two chlorites coming in 20–30 % *high*. A thin film cannot be the explanation of the absolute deficit and be absent from the ratios at the same time, and the ratios are the better measurement of the two.

What remains, and what would distinguish them:

### The irradiated length

The beam-overflow correction (§2.10) is applied with an assumed 20 mm of specimen, for both the clays and the corundum. With a fixed 0.5° slit the beam is 20 mm long at 6° and 30 mm at 4°, so a deposit shorter than the pressed standard loses intensity the correction does not restore. This can reach a factor of two, not four, but it is the one candidate that is a measurement rather than a hypothesis: *the physical length of a clay mount's deposit along the beam, against the corundum's*.

### The deposit's mass per unit area

If the mounts are weighed before and after deposition, the mass in the beam is known outright and  $k_p$  stops being inferred from a purity at all — (2.28) can be replaced by a direct comparison, and the thickness question above is answered by the same number.

### A common factor in the basal calculation

Anything that multiplies a whole basal series by one constant per gram, independent of the mineral, is indistinguishable from the above on this data. It would have to leave the ratios of §A.17 untouched, which most candidates do not.

## Why $r$ is fixed rather than fitted, restated with numbers

The same four mounts were run a second way, with the library's own interstratified entries at the pure end member — which carry the crystallite-thickness broadening a powder calculation lacks, and fit far better — leaving  $r$  free:

| Mount             | $R_{wp}$ , powder | $R_{wp}$ , library | $k_p$ , $r$ fixed | $k_p$ , $r$ free |
|-------------------|-------------------|--------------------|-------------------|------------------|
| Kaolinite         | 71 %              | 71 %               | 0.245             | 0.245            |
| Chlorite A        | 94 %              | 56 %               | 0.216             | 0.391            |
| Chlorite B        | 94 %              | 40 %               | 0.281             | 0.292            |
| Illite            | 86 %              | 54 %               | 0.247             | 0.100            |
| Ratio of extremes |                   |                    | 1.30              | 3.90             |

The better-fitting route gives the worse calibration, by a factor of three in spread. That is the  $r$ -degeneracy of §A.14 measured rather than argued: a basal-dominated pattern does not determine  $r$ , so a fit free to choose it will trade  $r$  against scale at no cost in  $R_{wp}$  and carry the trade straight into the mass. A lower residual is not a better calibration, and this is the clearest case of it in the manual.  $r$  belongs to the hkl reflections, and a calibration should be made where they are.

## What this does not license

**$k_p$  remains 1 in the code.** Four mounts, one preparation, one operator, one instrument and one standard is a result, not a calibration table, and the factor is an instrument constant that no other laboratory could adopt. Beyond that:

- Every mount is air-dried. The quantitative fit is made on the glycol-solvated mount (§2.1), so none of this bears on the expandable clays, which are the phases whose texture is least like a chlorite's.
- The purities used (0.81 to 1.00) come from the accompanying phases in the same fits, so they carry the same modelling error they are meant to correct. The illite's 0.81 is the weakest of them.
- The powder-route residuals are 71–94 %. A three-dimensional powder calculation with one orientation parameter is a poor description of an oriented mount, and these masses inherit that. The constancy of the factor across four such fits is more informative than any one of them.
- Smectite has no pure mount here at all, and no calibration.

What it does establish: the machinery is complete and checked — absorption coefficients against published compounds, the conditions of two measurements compared rather than assumed, the infinite-thickness assumption computed rather than hoped for — and applied to real data it returns one factor, not four unrelated ones. When the glycol and heated mounts are measured, the same three commands give  $k_p$  for those, and the question of whether it is one factor per preparation or one per mineral will be answerable.

## A.13 Summary

| Area                                  | Tests                                                                                                                                           | Status                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Interstratification model             | 25 automated, against closed-form and invariant results                                                                                         | Pass                                                                      |
| Basal / three-dimensional consistency | 9 automated, four phases                                                                                                                        | Pass                                                                      |
| External validation                   | 4 automated, against Reynolds (1965); $R = 0.039$ vs his 0.042                                                                                  | Pass                                                                      |
| Workflow on synthetic data            | 7 automated                                                                                                                                     | Pass                                                                      |
| Structure import                      | 209 phases; 12 positions to $0.03^\circ$ ; origin convention established; every phase checked against the cell mass and volume its block states | Pass; one incomplete structure found                                      |
| File readers                          | 33 files; bit-for-bit against vendor export                                                                                                     | Pass                                                                      |
| Zero error                            | 11 specimens $\times$ 3 mounts                                                                                                                  | Pass after correction                                                     |
| Mineral search                        | 4 methods compared on known and real data                                                                                                       | Method 4 adopted                                                          |
| Peak stripping                        | 5 widths, with and without peaks, against a known background                                                                                    | Defect found and corrected; 12 automated                                  |
| Weight percent                        | 11 automated, against known masses                                                                                                              | Exact; needs a texture calibration                                        |
| Internal standard                     | 10 automated, on both bases                                                                                                                     | Pass; the absolute scale and the unaccounted fraction                     |
| Interstratification reporting         | 5 automated, across the I/S series                                                                                                              | Host-layer fraction reported beside the phase name                        |
| Parameters on a range edge            | 12 automated, including that the end member is the pure phase and that a physical limit reads differently from a library one                    | Reported by every fit; both series extended to their end member           |
| Pure-mineral basal ratios             | 14 automated; three single-phase mounts; eight candidate causes of the chlorite shortfall                                                       | Kaolinite 002/001 = 1.05; one conclusion retracted; the chlorite 004 open |
| Absorption                            | 11 automated; six compound coefficients against published values; the Cu edge discontinuity                                                     | Corundum 31.14 vs 31.1; an unlisted element raises                        |
| External standard                     | 10 automated; four pure mounts against a corundum on the same settings                                                                          | $k_p$ 0.216–0.281, 11 % relative sd over $r^{-3} \times 244$              |
| Refined structures                    | 27 automated; against a refinement of one specimen, and against a CIF written by TOPAS                                                          | $R_{wp}$ 35.9 $\rightarrow$ 31.2 %; adopted for the clays                 |
| Quantitative fit                      | 11 real specimens, after the correction                                                                                                         | $R_{wp}$ 37–68 %, median 45 %                                             |
| File pairing                          | 12 automated, on real naming conventions                                                                                                        | Pass                                                                      |
| Names used but never imported         | every source file, checked with pyflakes                                                                                                        | 3 found in the interface and fixed                                        |
| Interface                             | End-to-end in a browser, every tab and button                                                                                                   | Pass; 7 defects found and fixed                                           |

The automated suite comprises **258 tests** and runs in about eighty seconds (`pytest -q`). Those that need the ICSD structures are skipped automatically when they are not installed; the rest, including the whole of `tests/test_background.py`, run without them.

## Appendix D — Provenance of the accompanying-mineral structures

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The 209 structures below are those of the TOPAS structure library used during development. They are **not** distributed with ClayQuant and are not part of it: the library is user-supplied, imported by `clayquant-import-structures` (§4.3), and every ClayQuant installation may use a different one. The table is given so that any pattern the program calculates can be traced to the structure it came from, and that structure to its source database and entry.

**Why the citations are not printed here.** Of the 209 entries, 200 are ICSD, 3 AMCSD and 2 COD, and 4 carry no identifier. ICSD is a licensed database with no open lookup from an entry number to its publication, so those citations could not be retrieved and verified for this manual, and they have deliberately *not* been guessed: an invented reference is worse than an absent one. The identifiers below are exact, taken from the library file itself, and each resolves to its full bibliographic record in the database named — through an ICSD subscription, through *rruff.geo.arizona.edu/AMS* for AMCSD, or through *crystallography.net* for COD, all of which report the original publication for the entry. A machine-readable copy with an empty `citation` column to fill in is in the repository as `docs/structure_provenance.csv`.

The four clay mineral structures are not in this table: they are integral to the program, are named in §4.2, and their publications are cited in full in §10.

| Phase          | Database | Entry  |
|----------------|----------|--------|
| Actinolite     | ICSD     | 68319  |
| Aegirine       | ICSD     | 156560 |
| Akermanite     | ICSD     | 158177 |
| Albite         | ICSD     | 68913  |
| AliteM3        | ICSD     | 94742  |
| Almandine      | ICSD     | 71895  |
| AluminateC     | ICSD     | 1841   |
| AluminateO     | ICSD     | 1880   |
| Aluminium      | ICSD     | 606001 |
| Amorphous      | —        | —      |
| Analcime       | ICSD     | 156593 |
| Anatase        | ICSD     | 9852   |
| Andalusite     | ICSD     | 84613  |
| Anglesite      | ICSD     | 92609  |
| Anhydrite      | ICSD     | 16382  |
| Ankerite       | ICSD     | 66338  |
| Anorthite      | ICSD     | 34942  |
| Apatite        | ICSD     | 26204  |
| Aphthitalite   | ICSD     | 26018  |
| Aragonite      | ICSD     | 157992 |
| Arcanite       | ICSD     | 2827   |
| Augite         | ICSD     | 90143  |
| Austenite      | ICSD     | 186833 |
| Azurite        | ICSD     | 158577 |
| Baricite       | ICSD     | 202099 |
| Barite         | ICSD     | 16904  |
| Bassanite      | ICSD     | 79529  |
| Bayerite       | ICSD     | 26830  |
| Belite Alpha H | ICSD     | 82997  |
| Belite Beta    | ICSD     | 81096  |
| Biotite        | ICSD     | 95359  |
| Boehmite       | ICSD     | 41215  |
| Bornite        | ICSD     | 1963   |

|                                               |       |        |
|-----------------------------------------------|-------|--------|
| Bredigite                                     | ICSD  | 9828   |
| Brookite                                      | ICSD  | 36408  |
| Brownmillerite                                | ICSD  | 2842   |
| Brucite                                       | ICSD  | 79031  |
| Brushite                                      | ICSD  | 16132  |
| Ca <sub>2</sub> P <sub>2</sub> O <sub>7</sub> | ICSD  | 14313  |
| Calciolangbeinite                             | ICSD  | 290622 |
| Calcite                                       | ICSD  | 73446  |
| Calcite-Superlorentz                          | ICSD  | 73446  |
| Caledonite                                    | ICSD  | 164674 |
| Cassiterite                                   | ICSD  | 262768 |
| Celestine                                     | ICSD  | 22322  |
| Cementite                                     | AMCSD | 13523  |
| Cerussite                                     | ICSD  | 6178   |
| Chalcopyrite                                  | ICSD  | 2518   |
| Chamosite                                     | ICSD  | 80080  |
| Chlorite                                      | ICSD  | 164234 |
| Chlorite Pawley                               | ICSD  | 164234 |
| Chloroxiphite                                 | ICSD  | 164137 |
| Chromite                                      | ICSD  | 171121 |
| Chrysotile                                    | ICSD  | 77747  |
| Coesite                                       | ICSD  | 100749 |
| Copper                                        | ICSD  | 53247  |
| Corundum                                      | ICSD  | 51687  |
| Cotunnite                                     | ICSD  | 43344  |
| Covellite                                     | ICSD  | 63327  |
| Cristobalite                                  | ICSD  | 75483  |
| Cubanite                                      | ICSD  | 42104  |
| Cuprite                                       | ICSD  | 63281  |
| Diaboleite                                    | ICSD  | 15382  |
| Diamond                                       | ICSD  | 52054  |
| Diaspore                                      | ICSD  | 29344  |
| Dickite                                       | ICSD  | 98132  |
| Diopside                                      | ICSD  | 17043  |
| Dolomite                                      | ICSD  | 100680 |

|                          |       |        |
|--------------------------|-------|--------|
| Dolomite-Ankerite SS     | ICSD  | 100680 |
| Dolomite-Superlorentz    | ICSD  | 100680 |
| Ellestadite              | ICSD  | 262245 |
| Enstatite                | ICSD  | 9328   |
| Enstatite-Ferrosilite SS | ICSD  | 9328   |
| Epidote                  | ICSD  | 168464 |
| Epsomite                 | ICSD  | 16595  |
| Eskolaite                | ICSD  | 75577  |
| Ettringite               | ICSD  | 155395 |
| Faujasite                | ICSD  | 40931  |
| Fayalite                 | ICSD  | 54029  |
| Ferrosilite              | ICSD  | 34858  |
| Fluorite                 | ICSD  | 60559  |
| Forsterite               | ICSD  | 9685   |
| Forsterite-Fayalite SS   | ICSD  | 9685   |
| Franklinite              | ICSD  | 91827  |
| Galena                   | ICSD  | 63095  |
| Gehlenite                | ICSD  | 158171 |
| Ghiaraite                | AMCSD | 19787  |
| Gibbsite                 | ICSD  | 184708 |
| Goethite                 | ICSD  | 245057 |
| Goslarite                | ICSD  | 107287 |
| Graphite                 | ICSD  | 76767  |
| Griceite                 | ICSD  | 41409  |
| Grossular                | ICSD  | 83461  |
| Grunerite                | ICSD  | 24214  |
| Gypsum                   | ICSD  | 151692 |
| Halite                   | ICSD  | 181148 |
| Hematite                 | ICSD  | 40142  |
| Hercynite                | ICSD  | 40093  |
| Heulandite               | ICSD  | 37064  |
| Hornblende               | ICSD  | 158242 |
| Hydrocalumite            | ICSD  | 51890  |
| Hydrocerussite           | ICSD  | 280932 |
| Hydrotalcite             | ICSD  | 81963  |

|                                  |      |         |
|----------------------------------|------|---------|
| Illite                           | ICSD | 90144   |
| Illite Pawley                    | ICSD | 90144   |
| Ilmenite                         | ICSD | 30664   |
| Iron                             | ICSD | 76747   |
| Jarosite                         | ICSD | 166799  |
| Jasmondite                       | ICSD | 26407   |
| K <sub>2</sub> ZnCl <sub>4</sub> | —    | —       |
| Kaolinite 1M                     | ICSD | 63192   |
| Kaolinite 2M                     | —    | —       |
| Kieserite                        | ICSD | 68345   |
| LaB <sub>6</sub>                 | ICSD | 248440  |
| Labradorite                      | ICSD | 100235  |
| Langite                          | ICSD | 30724   |
| Laurionite                       | ICSD | 28035   |
| Lead                             | ICSD | 96501   |
| Lepidocrocite                    | ICSD | 108876  |
| Leucite                          | ICSD | 161630  |
| Lime                             | ICSD | 163628  |
| Litharge                         | COD  | 9012700 |
| Lizardite                        | ICSD | 202358  |
| Maghemite                        | ICSD | 79196   |
| Magnesioferrite                  | ICSD | 41290   |
| Magnesite                        | ICSD | 73731   |
| Magnesite-Siderite               | ICSD | 73731   |
| Magnetite                        | ICSD | 158504  |
| Malachite                        | ICSD | 262794  |
| Marcasite                        | ICSD | 42415   |
| Massicot                         | ICSD | 15403   |
| Mayenite                         | ICSD | 241243  |
| MCPM                             | ICSD | 133     |
| Melanterite                      | ICSD | 157296  |
| Mendipite                        | ICSD | 93003   |
| Merwinite                        | ICSD | 26002   |
| Microcline                       | ICSD | 83531   |
| Minium                           | ICSD | 42484   |

|                                 |       |        |
|---------------------------------|-------|--------|
| Monetite                        | ICSD  | 2745   |
| Montmorillonite                 | ICSD  | 51636  |
| Mullite                         | ICSD  | 75305  |
| Muscovite                       | ICSD  | 180081 |
| Natron                          | ICSD  | 97922  |
| Newberyite                      | ICSD  | 31281  |
| Nordstrandite                   | ICSD  | 16018  |
| Oldhamite                       | ICSD  | 30236  |
| Orthoclase                      | AMCSD | 20540  |
| Otavite                         | ICSD  | 156740 |
| Paralaurionite                  | ICSD  | 74291  |
| PbS <sub>2</sub> O <sub>3</sub> | ICSD  | 69532  |
| PbSO <sub>3</sub>               | ICSD  | 68889  |
| Periclase                       | ICSD  | 159367 |
| Perovskite                      | ICSD  | 62149  |
| Phosgenite                      | ICSD  | 4240   |
| Pigeonite-C                     | ICSD  | 9248   |
| Pigeonite-P                     | ICSD  | 9247   |
| Portlandite                     | ICSD  | 64950  |
| Prehnite                        | ICSD  | 164238 |
| Pseudoboleite                   | ICSD  | 67680  |
| Pumpellyite                     | ICSD  | 158186 |
| Pyrite                          | ICSD  | 15012  |
| Pyrope                          | ICSD  | 71892  |
| Quartz                          | ICSD  | 67121  |
| Qusongite                       | ICSD  | 246149 |
| Riebeckite                      | ICSD  | 200423 |
| Rozenite                        | ICSD  | 23912  |
| Rutile                          | ICSD  | 36412  |
| Sanidine                        | ICSD  | 90141  |
| Scotlandite                     | ICSD  | 30993  |
| Siderite                        | ICSD  | 100678 |
| Silicon                         | ICSD  | 51688  |
| Sillimanite                     | ICSD  | 66330  |
| Smithsonite                     | ICSD  | 100679 |

|                                                                  |      |         |
|------------------------------------------------------------------|------|---------|
| Spessartine                                                      | ICSD | 83458   |
| Sphalerite                                                       | ICSD | 77090   |
| Spinel                                                           | ICSD | 77583   |
| Spurrite                                                         | ICSD | 156625  |
| Staurolite                                                       | ICSD | 53148   |
| Stilbite                                                         | ICSD | 83469   |
| Stishovite                                                       | ICSD | 40098   |
| Sylvite                                                          | ICSD | 165593  |
| Syngenite                                                        | ICSD | 157072  |
| Taenite                                                          | ICSD | 103555  |
| Talc                                                             | ICSD | 100682  |
| TCP Alpha                                                        | ICSD | 923     |
| Tenite-Superstructure                                            | ICSD | 56386   |
| Tenorite                                                         | ICSD | 67850   |
| Thaumasite                                                       | ICSD | 185680  |
| Titanite                                                         | ICSD | 50144   |
| Tobermorite11A                                                   | ICSD | 92942   |
| Tridymite                                                        | ICSD | 34867   |
| Troilite                                                         | ICSD | 35005   |
| Trona                                                            | ICSD | 192710  |
| Vaterite                                                         | ICSD | 181959  |
| Vermiculite                                                      | ICSD | 192347  |
| Vivianite                                                        | ICSD | 423390  |
| Wadalite                                                         | ICSD | 151561  |
| Weddellite                                                       | ICSD | 192661  |
| Whewellite                                                       | ICSD | 153499  |
| Whitlockite (Ca,Mg) <sub>3</sub> (PO <sub>4</sub> ) <sub>2</sub> | ICSD | 1490    |
| Whitlockite Ca <sub>3</sub> (PO <sub>4</sub> ) <sub>2</sub>      | —    | —       |
| Wollastonite                                                     | ICSD | 201538  |
| Wuelfingite                                                      | ICSD | 50447   |
| Wuestite                                                         | ICSD | 27237   |
| Wurtzite                                                         | ICSD | 67453   |
| Yeelimite                                                        | COD  | 4001772 |
| Zincite                                                          | ICSD | 154486  |
| Zinkosite                                                        | ICSD | 71018   |

|        |      |      |
|--------|------|------|
| Zircon | ICSD | 9582 |
|--------|------|------|

Entries without an identifier are either not structural models (the amorphous and mixed-layer entries) or were added to the library without a database reference. Two entries of the library are Pawley (whole-pattern, structureless) phases for illite and chlorite; they carry the ICSD numbers of the structures their cells were taken from, which are the same illite and chlorite structures ClayQuant itself uses.

## Appendix B — File formats read

| Format                  | Extension             | Notes                                                                                                                                                                                                                                                  |
|-------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PANalytical XRDML       | .xrdml                | Intensities from <counts> or <intensities>; multiple appended scans are concatenated; positions from a start/end range or an explicit list. Wavelengths, goniometer radius, divergence slit, counting time and sample identifier are read as metadata. |
| Two-column text         | .xy, .txt, .dat, .asc | 2θ and intensity. Comment and header lines are skipped; comma, tab and space separation are accepted.                                                                                                                                                  |
| Bruker RAW              | .raw                  | Binary; the common RAW1.01 and RAW4 variants. Support is best-effort — if a file is not read, export it as two-column text from the vendor software.                                                                                                   |
| CIF                     | .cif                  | The subset produced by ICSD, COD and AMCSD exports: cell, an explicit symmetry-operation loop, and an atom-site loop with occupancies and $B_{\text{iso}}$ or $U_{\text{iso}}$ .                                                                       |
| TOPAS structure library | .xml                  | jEdit macro-menu form; str blocks with cell, space group and site lines.                                                                                                                                                                               |
| Pattern library         | .npz                  | Compressed NumPy archive holding the grid, the pattern matrix, and per-entry and library metadata.                                                                                                                                                     |

# Appendix C — Symbols

| Symbol                  | Meaning                                                                      | Units             |
|-------------------------|------------------------------------------------------------------------------|-------------------|
| $s$                     | scattering vector length, $2 \sin\theta/\lambda = 1/d$                       | $\text{\AA}^{-1}$ |
| $q$                     | $\sin\theta/\lambda = s/2$ ; argument of scattering and displacement factors | $\text{\AA}^{-1}$ |
| $d$                     | interplanar spacing; $d_A$ , $d_B$ layer repeat distances                    | $\text{\AA}$      |
| $F(s)$                  | layer structure factor, complex                                              | electrons         |
| $B$                     | isotropic displacement parameter                                             | $\text{\AA}^2$    |
| $N, \langle N \rangle$  | layers per crystallite; its mean over the CSDS                               | —                 |
| $\beta$                 | width of the lognormal CSDS in $\ln N$                                       | —                 |
| $G(n)$                  | pair weight at layer separation $n$                                          | —                 |
| $\mathbf{P}, P_{mn}$    | Markov transition matrix and its elements                                    | —                 |
| $\mathbf{W}, W_m$       | stationary layer proportions                                                 | —                 |
| $\mathbf{T}$            | phase-weighted transition matrix, equation (2.10)                            | —                 |
| $R$                     | Reichweite, the range of the stacking correlation                            | —                 |
| $r$                     | March–Dollase preferred-orientation parameter                                | —                 |
| $\alpha$                | angle between a scattering vector and the texture axis $c^*$                 | rad               |
| $\gamma$                | equatorial divergence of the incident beam                                   | rad               |
| $R_{\text{goniometer}}$ | goniometer radius                                                            | mm                |
| $c_p$                   | fitted scale factor of library pattern $p$                                   | —                 |
| $w_i$                   | statistical weight of point $i$                                              | counts $^{-1}$    |
| $R_{\text{wp}}, R_p$    | weighted-profile and profile agreement factors                               | —                 |

ClayQuant 0.1.0 — user and reference manual. The program is distributed under the MIT licence. Structure data is not distributed with it and remains subject to the licence of its source. This manual documents the state of the program at the revision noted on the title page; the test record of Appendix A refers to that revision.